

TESLA

New ideas in sensational article by
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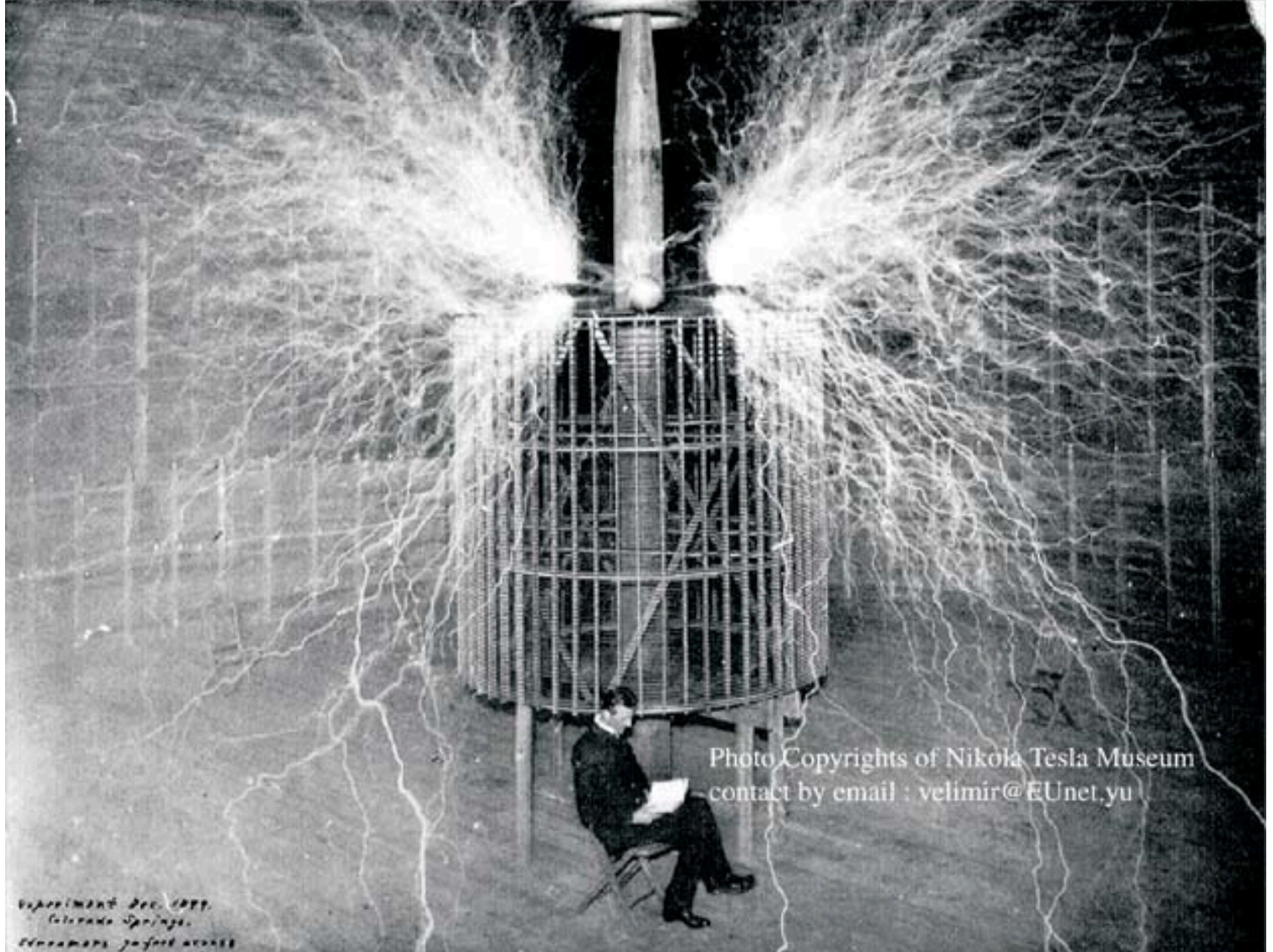


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Tesla

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(Editor's notes made by Alexander V. Frolov)

*"Things were created rather in the minds of angels, than in Nature, i.e. angels had all things clear in their minds and thoughts before they got a real existence."
Words by Augustine Blessed cited by
H.P. Blavatskaya in her book "Secret doctrine"*

"I will not work for the present no more, I am working for the future"- said Tesla to journalists in New York City more than seven decades ago. – "The future is mine!" Inventor of alternating current, polyphase motor-generator, reversible magnetic field, radio, teleautomatics... Tesla was inventor, on whose patents energetics of XX century is based, he worked alone on explanation of cosmic processes during decades, having a desire to join material and spiritual theoretically like he did it in his practical discoveries.

Mentioning of Nikola Tesla's name now mainly connected with the so-called Tesla coil, inductive motor and international unit symbol to measure the force of magnetic field. Many facts of his life and his extraordinary creative gift are forgotten.

The most productive period of his creative work Tesla spent in USA. He had patented more than 300 inventions in different countries. Many of them still were not repeated; for example, his receiver of radiant energy. We know nothing definite about its principle of operation besides that may be it is the question of **transformer of cosmic rays energy**. In the period from 1899 until 1900, he devoted himself to the research of electromagnetic oscillations of very low frequencies in the specially built laboratory in Colorado Springs. Two years later he began to build a World Translation Station on Long Island near New York and he didn't finish it. Tesla's friend American steel magnate J. P. Morgan financed this experiment. After this project was closed in 1905, Tesla up to his death preferred to work alone, far from people's sight. But this period was not without new discoveries. Just at that time being a mature scientist he came to fundamental conclusions, which should become a new milestone in future science. We know from history that when scientific thought finds itself at crossroads, the scientists address to the past seeking for support and inspiration. Let's try to answer many important questions.

How Tesla came to his discoveries? These discoveries are: influence of super low-frequency electromagnetic waves on biological systems, especially on the work of brain, merging of energy structures, the so called "fire balls", which produced by inductive fields of primary and secondary electromagnetic coils, superconductivity

of natural and artificial mediums, the so called wireless energy transfer and so on.

What are the main axioms of Tesla's cosmology? How do they follow from his metaphysics? How did he apply them in his physical experiments? Why theorists and empiricists of modern physics of time are so interested in reconstruction of Tesla's theory of physical reality and his view on electromagnetic phenomena? Why Tesla didn't formulate his scientific theory and didn't publish it? Can Tesla's sight on ethic aspect of scientific discoveries help to refine modern natural sciences, especially physics, which is in crisis of ideas now? What can we expect from studying of Tesla's conceptions in near future? Whether it will be an overestimate to say that Tesla in 1900 grounded the possibility of global information-oriented society in his famous project "World-Wide System"? Whether it is actually technical and technological basis of that, we call now "New World Order"? Shall we consider Tesla to be a spiritual precursor of a new scientific and technological civilization named as Tesliana, where the **technology of "time design" is prevalent and only and inexhaustible energy source is the asynchronicity of various levels of physical processes**.

*(Editor's note: Time as energy source is the most important topic of many research efforts. How can we explain this way in simple words? To my mind it is difference in energy of oscillations that is formulated as $E=hn$, where n is frequency, that depends on scale of space, i.e. speed of time course. Let's assume that some system of energy E can be moved from space A in local space of accelerated time B . In projection to the space A this means increase of energy since the same wavelength in space B should be less. Let's assume some "bridge" or "energy exchange channel" between space A and space B . In this case we'll be able to create some energy exchange between A and B due to **natural gradient of energy density**. Prof. Abramovic wrote about "asynchronicity of various levels of physical processes" but more accurate notion here is **"relative difference in the speed of time"**, which connected with speed of light and physical properties of space. "Asynchronicity" can be detected for two oscillation processes of the same frequency and to my mind in this case it cannot be used to produce the gradient of energy density).*

TESLA'S RESEARCH METHOD

Let's return to the middle XIX century, to the little village Smilyan in Lika, which was Austrian-Hungarian province. In this village on July 10, 1856, Serbian Orthodox priest Milutin Tesla saw the birth of his forth child Nikola. Tesla's family didn't allow him to study in Polytechnic Institute. His father required him to be a priest. Nikola felt a deep and insatiable vocation to be an electrical engineer and he felt sick due to this disagreement. When crisis came and it became evident that he can die, father finally agreed with desire of his son. Soon Tesla recovered as if it was a miracle and was absorbed in inventive fantasy. After mental efforts

he began to suffer from a strange phenomenon. He saw clear visions sometimes accompanied by strong light flash. It is the ability of people who have parapsychological power - "In my boyhood I suffered from a peculiar affliction due to the appearance of images, often accompanied by strong flashes of light, which covered the sight of real objects and interfered with my thought and actions... When a word was spoken to me the image of the object it designated would present itself vividly to my vision and sometimes I was quite unable to distinguish whether what I saw was tangible or not."

He also described his spontaneous mind traveling: "Then, instinctively, I commenced making mind trips beyond the limits of the small world of which I had knowledge and I saw new scenes. These were at first very blurred and indistinct, and would flit away when I tried to concentrate my attention upon them but, by and by, I succeeded in fixing them; they gained in strength and distinctness and finally assumed the concreteness of real thing. I soon discovered that my best comfort was attained if I simply went on in my vision further and further, getting new impressions all the time, and so I began to travel - of course in my mind. Every night (and sometimes during the day) when alone, I would start on my journeys, see new places, cities and countries, live there, meet people and make friendships and acquaintances and, however unbelievable, it is a fact that they were just as real and dear to me as those in actual life and not one bit less intense in their manifestations. This I did constantly until I was seventeen, when my thoughts turned seriously to invention."

Tesla was satisfied to notice that he could clearly visualize his discoveries; **he even didn't need experiments, models and drawings. In such a way he developed his own method to give material form to creative ideas.** Tesla sharply distinguished ideas, which were coming into his mind as visions and those, which appeared due to analytical mind clearance. Tesla explained: "The moment when somebody designs an imaginary device is connected with a problem of transfer from raw idea to the practice. That's why any discovery made this way has lack of details, and usually it is inferior... My method is another one." He did not hurry up with empirical check. When idea appeared, he immediately began to elaborate it in his imagination. He would change its design, improving it and would "switch on" the device to begin its operation in his head. For him it did not matter whether he tested his invention in the laboratory or in his mind. He even had time to notice if something prevents his invention to work properly... The same way he could develop a perfect idea without any touch. Only then he would give a concrete look for this final product of his mind. All his inventions were developed by this method with no exceptions during twenty years... There is hardly any scientific discovery that can be foresaw only mathematically, without visualization... Introduction of unfinished, crude ideas to the practice is always a waste of energy and time.

Studying the mechanisms of his mental life, Tesla found that a number of visions "from virtual reality" always have certain connection with events of "true reality". Soon he got an ability to realize this causation. He was glad to understand that every his thought is a result of influence of external impressions. He noted that not only thoughts but actions also appear in the same way. Some time passed and it became clear to him that he was only a kind of "automaton of cosmic forces" endowed with abilities to move, responsible on irritation of sense organs and thoughts. Invention of teleautomatic control was a result of this conclusion many years later. He understood the laws of this control through firstly unclear feeling of external conditioning of his own behavior.

Nikola Tesla used imagination as a psychological precursor of well-ordered disclosure made by mathematical methods. We can say nothing of this kind about his contemporary inventor Thomas Edison, because he had "weak knowledge in mathematics, and he at first was guided by long and laborious experiments".

In his records, Tesla often speaks about his predisposition to the mental processes. Principle of these processes corresponds to the same principles, which Nature should follow. He consider this inborn gift to be given as a "diffused pressure", which calls a sense to make another invention, because the experiment of previous researcher was short of something. Here he sees not only a source of invention in general, but also some proof that the law of different reality affects the man. To be short, Tesla considers creative imagination to be a threshold of conscious discovery act.

According to his words, the certain conclusions were born in his mind spontaneously, **in the form of geometrical images.** Then followed the understanding of principle of discovery and its physical interpretation. Only then mathematical formalization took place followed by finding of necessary technical properties of material essential for continuous operation of constructed physical model. The work on inventions he understood at first as a struggle for mental clarification, i.e. elimination of secondary ideas and sensible details, which blur imaging principle and complicate the approach to the real nature of relations between fundamental geometrical elements.

At the same time when Crookes began scientific studying of spiritistic phenomena in London, Mendeleev began to do the same in Saint Petersburg in 70-s of XIX century. Committee of specialists formed in Saint Petersburg after a short work (about ten experiments) concluded that it is just a superstition.

Since then in England and Russia science is subdivided on secret, "fictitious" including Tesla's physics of ether and clear, "real", university science of commercial and even political direction, for example, nuclear physics and theory of relativity.

TESLA AS A PHYSICIST

In theoretical and experimental physics of XX century, we can clearly define three different styles of thinking. Quantum mechanics as well as relativism (near-light speeds) and traditionalism to which Tesla belonged (really he is a non-recognized scientist of the future) tried to find truth, i.e. real nature of time, space and especially the essence of motion.

Before we'll go into Tesla's metaphysics, namely into the world of fundamental processes of his physics, let's try to pinpoint the place, which Tesla ranks in the world of modern scientific systems.

Planck's constant, Heisenberg's uncertainty relation, Pauli principle and wave function by Shredinger were the main theoretical instruments of quantum and mechanical approach to the cosmic phenomena. The main goal of quantum mechanics, which is to discover fundamental or elementary particle of matter, is still not realized. In spite of great success in studying of process of chain reactions activation and physics of atomic nucleus, processes of formation of nuclei of heavy elements from nuclei of light elements, many things are still mysterious and they are beyond technical control. **Starting from tendency of general development, we can make a conclusion that productive basis of notions of quantum-mechanical physics is exhausted.** It became evident that we should research such categories as time and space again, at deeper level of realization, at ontological level to determine a true nature of physical processes.

Special theory of relativity by Albert Einstein published in 1905 in German magazine "Annalen der Physik" marked the second milestone in theoretical physics and used four-dimensional space-time continuum by Minkovsky as a model of physical reality.

The third method to understand physical reality relates to Tesla's ideas. **Starting from works by M. Faraday and D. Arago from the one hand and L. Galvani and A. Volta from the other hand, Tesla could create his own original theory of world on the basis of theory of acoustic resonators by Helmholtz and modified ether model by Lord Kelvin. This theory gave outstanding results in experiments. Initial axiom of his theory was that universal energy of one physical system is based on the laws of resonance of vibrations, on coincidence of oscillations of system's parts. He considered that theory of ether cannot be excluded from physics, because matter and space cannot at all be completely and strictly separated.**

(Editor's notes: You can find this simple and important aspect in the letter of M. Faraday to R. Taylor. It is Faraday's opinion about the nature of matter. In his letter to Richard Taylor of June 25, 1844, M. Faraday wrote: So, matter anywhere is continuous... it is not necessary to assume the difference between atoms of matter and some intermediate space. The forces around centers provide these centers with properties of matter... M. Faraday

considered this aspect to make a conclusion for the nature of electric conductivity and insulation. However, his point of view can be developed in gravitational aspect also. Atom and space around of an atom (physical vacuum) is the same ether matter in different states. In some articles it was demonstrated that this difference can be defined as opposite directions of the time flow for "inner cosmos" of matter and "spatial cosmos" between atoms of matter. Energy balance is provided by this way in the Universe.)

It is Tesla's opinion that electrified state is a fluid state based on the substance having the properties of perception and elements of consciousness. In mathematics, the scientist was the follower of realistic approach, holding the idea the properties of mathematical and physical objects directly correspond to each other. In experimental approach, Greek mechanician Archimedes served as an example to him. Archimedes stated that "time should be excluded from physics" as an excess phenomenon. It should be noted that even in XIX century scientists H. Hertz and D'Alembert tried to create theoretical physics without notion of force, but this attempt failed. Russian scientist N.A. Kozyrev followed it, but his theory also didn't achieve the level of practical application. **Problem of relation between force and time in physics is still unsettled.**

Tesla's experiments mainly consist in ascertainment of deep common properties of physical systems, which should be brought to resonance. Evidence to it was his complex electromagnetic oscillator, Wardencliff Tower (built at Long Island near New York in 1901-1905). He could make simultaneous vibrations of ionosphere and the globe by means of this tower.

In mathematics as well as in physics, Tesla was the follower of strict determinism. He held sacred mathematics, but also considered that physical processes could be described not only by mathematical method. It is Tesla's opinion that control of process shall be made by foresight; just the foresight provides control. Such position is radically different from the theory of relativity. According to this theory, objective (immediate) cognition is impossible and reality is recognized by mathematical calculation.

Views of Tesla and Einstein on the problem of physical reality are fundamentally different. According to Einstein, human experience is relative, fictitious and doesn't correspond to true physical reality. For Tesla physical reality is universal and penetrates all levels of cosmic existence, i.e. nobody in any way can avoid certain cognition of the truth. According to Einstein, Ether is not a real category, but exists as a result of fallacious scientific opinions. For Tesla Ether is a unified undifferentiated field, which consists of time, space, energy, and formation of matter is a result of resonant processes in ether.

According to Einstein, Time is only a rank of phenomena, it is not a substantial physical entity or even category and it is registered only by means of measurements with

reference to every system. For Tesla Time is a real algorithm of materialized Mathematics and it is created from Ether due to resonance of physical systems, and it returns to Ether.

(Editor's note: Detailed conception on correlation between notions of Time and Ether you can read in articles by Dr. Belostotsky, Dr. Mishin and others, for example, New Energy Technolgies #3, p.15.

*In short: **Control on time course can be designed as control on ether density and direction of the ether wind. Technically it can be realized with longitudinal electromagnetic waves and Tesla worked on development of this special equipment.***

According to Einstein, maximal speed is achieved in Vacuum, it is a speed of light equal to 300 000 km/sec. For Tesla speed of electromagnetic waves is not limited, and experiments and calculations were made and showed that transfer of waves and energy at any distance is possible in principle and speed of mechanical and electrical waves penetrating the Earth highly exceeds speed of light in vacuum.

Speaking to his friends, Tesla often refuted some of Einstein's statements, especially those, which were related with curvature of space. He considered that it breaks the law of action and counteraction: "If curvature of space is formed due to the huge gravitational fields, than it should become straight due to the counteraction."

(Editor's note on this Tesla's argumentation: This Einstein's notion of curvature here cannot be considered as usual spatial 3-dimensional curvature of some material object, but it is 4-dimensional notion. If someone became aware of its 4-dimensional sense, then this curvature and 4-dimensional elasticity become really useful notions for investigators of time. Just this elasticity produces force, which moves matter in time from the Past to the Future.)

Einstein was a pure theorist and Tesla was mainly an experimenter. There are no information about contacts and talks between these two scientists. **Nevertheless, Einstein congratulated Tesla with his 75 anniversary. He underlined Tesla's most important merit in science, multiphase system of generators and alternating current motors, which is one of his least important discoveries.** Tesla believed: "**We shall have no need to transmit power at all. Ere many generations pass, our machinery will be driven by a power obtainable at any point of the universe... Throughout space there is energy...**"

TESLA'S VIEWS ON NEW WORLD ORDER

There is no doubt that Tesla had a great feeling of uniformity of terrestrial nature and cosmic processes. He brought the essence of his "World system" (announced in 1900) to 12 positions, which reflect the very essence of modern technical and technological network of world telecommunications. This system is

also a technological basis to establish global informational society. Tendency of this society to the universal unification manifests today in many things.

Tesla constructed his "World order" starting from principles of his own inventions. Let's enumerate some of them. **These are Tesla's transformer motors**, i.e. device for creation of electrovibrations with specific properties; **step-up transformer specially designed to excite the Earth** (this device with its electrical transmission means the same as a telescope in astronomy). **Then, Tesla's wireless system** including certain quantity of devices, which is now considered to be an unsurpassed method **to transmit electrical current without wires; a device for individualization of signal**, which in comparison with a primitive method to match signal means the same as advanced language in comparison with inarticulate mumbling. Due to this device, it is possible to transmit secret signal both in passive and in active form, because they do not interfere other transmissions and cannot be jammed by other signals. Any transmission is unique in its rhythm and has individual features; actually, it is an unlimited quantity of stations and instruments, which can act on one or two frequencies and at the same time they would not call any mutual jamming. **One more invention is wave processes in Earth's ionosphere;** in popular explanation this discovery means that **field of the Earth reacts on electrical vibrations of given frequency in the same way as a tuning fork resonates with a certain length of wave.** These electrical vibrations can have a strong influence on the field of the Earth. Such resonant effect can be used by different ways, and first to change universal energy system, which is far from to be perfect and didn't use up all its natural abilities.

Tesla defines the goals of "World Wide System" in such a way that it becomes clear: Establishment of dense communicational network of all kinds and levels should serve to humanize scientific technology. In this sense in the very beginning of XX century he was a prophet of XXI century, since **real metaphysical problems began to penetrate theoretical physics only now together with the problems of cosmogony, origin of the Universe, time and space.** Therefore, even in far 1900 Tesla considered that the following things would be necessary (they are already realized now):

1. Establishment of connection between existing telegraph stations or centers all over the world;
2. Organizing of secret state telegraph service without a possibility of jamming (it was not realized due to the fact that inventions by Markoni were used instead of Tesla's patents on radio transmission. Markoni's inventions required a separate frequency for every synchronous transmission);
3. Establishment of connection between existing telephone centers or stations on the Earth;
4. United diffusion of common gazette news by means of telegraph and telephone;
5. Organizing of service with principles of "World Wide System" to transmit private information (it was realized by the worldwide computer web, Internet);

6. Establishment of interconnection of all telegraph systems in the world;
7. Uniform timing mark by means of clock, which mark seconds with an astronomic accuracy;
8. Transmission of signs, words, sounds and so on as well as typewritten and hand-written texts (it is an exact description of fax machine);
9. Creation of service to record music;
10. Organizing of worldwide service for needs of mercantile marine, which should help in navigation, in compassless voyages? It will help to determine location and speed, to prevent collisions and accidents and so on;
11. Introduction of worldwide printing system;
12. Reproduction of photos and all kinds of drawings or hand-written texts with a possibility to send them over the world.

Though development of modern science and technology mainly confirmed Tesla's oracular ideas, it didn't match with them in fundamental sense, namely **all forethought ideas Tesla wanted to realize by means of transmission through the Earth, but not from surrounding orbits**. Due to the same reasons, which made him to stand up against using of nuclear (atomic) energy dangerous for people because of their ignorance of real structure of matter, Tesla considered that any infringement of cosmic space, magnetic field of the Earth is an infringement of harmony inherent to natural laws. In addition, it is a consequence of negative influence of people's free will, which becomes destructive in absence of ethical component in science.

Free will can be creative only with kindness, which accompanies to the higher understanding and conscious selection of positive intentions. He considered that humankind living on the Earth should understand all kinds of natural alliance with the planet otherwise people will be deprived of it.

In 1892-1893 in London Tesla read his famous lectures in Royal scientific society of Great Britain. Together with above-mentioned experiments he demonstrated many others, for example, lighting of lamps by wireless transmission of energy. Really, he only placed a vacuum neon tube to the field of transformer and the lamp lighted. Due to these wonderful lectures ("Light and other high frequency phenomena"), Tesla gained a good reputation and got many friends among scientists - William Crookes, sir Oliver Lodge and J.J. Thompson - lord Kelvin.

When returned to USA in 1893, he was absorbed in the problems of radio techniques, telecontrol and wireless transmission of energy at big distances without losses. He made successful experiments with direct electromagnetic field of the Earth and wireless transmission of energy at Colorado Springs. Though his dairy of this period of work kept safe and it was published, he didn't advertise his most important scientific results. However, all achieved brought him to an idea of universal system of wireless transmission of energy and information. Planetary transmission should

be based on such extraordinary inventions as **Tesla's resonance transformer without an iron core and Tesla's Magnifying transformer. Tesla expected a lot from extending transformer. It was a particular transformer made for exiting of the Earth.** By means of this transformer he made very power electromagnetic discharge at Colorado Springs. This discharge exceeded atmospheric lighting and Tesla got electricity, which made white-hot more than two hundred lamps placed at miles around the laboratory. For Tesla it was an irrefutable proof that his concept of wireless transmission of energy (distant action) was true.

After experiments at Colorado Springs in June 1900, he returned to New York and prepared the first universal system of wireless transmission, Wardencllyffe Tower, which he successfully tested three years later. Technical specifications of Tesla's transmitting station now are still unknown, and we cannot state with certainty, what was its real destination...

After unusual night of July15, 1903 when he fired the sky not only above New York but above all vast area of Atlantic Ocean, in 1905 Tesla suddenly left his laboratory without a clear reason and left all things untouched. As we know, he never crossed the threshold of Wardencllyffe, never visited it and never appeared in this region. It was very strange, he didn't take from this laboratory any calculation, drawing or document, none paper.

Last time there were suppositions that Tesla was the man who called a strong and strange by its consequences explosion in Siberia in the region of Podkamennaya Tunguska in 1908! It was considered that this explosion could call a meteorite, but there was not still found not the faintest fragment of it. It is also true that Wardencllyffe Tower was good in technical sense and Tesla passed energy through the Earth and could accumulate and discharge simultaneously a huge quantity of energy at any place of the planet. It fully matched the results of his work of many years standing on studying of Earth's properties as a medium for propagation of electromagnetic waves.

MISSING THEORY

Tesla didn't give us his own physical theory, but with numerous experiments he created a basis for new - resonance understanding of electromagnetism. He considered that **World is a uniform continuous electromagnetic medium and matter is one of manifestation of organized electromagnetic oscillations described by mathematical algorithm. He considered that resonance law is the most general natural law, which eliminates time and distance, and that all relations between phenomena are established only by mean of various simple and complex resonances, i.e. consistent vibrations of physical systems, mainly of electromagnetic nature.**

Finally, instead of Newton's integrals, Leibniz's differentials and Maxwell's theory of field lying in his calculations Tesla used simple mathematics of Ancient

Greek mechanicians, at first Archimedes, making an analogy between mechanics and electromagnetism. It is not possible to fully evaluate the significance of such method of thinking, which directly points out the necessity of more full physical interpretation of elementary mathematical notions.

Obviously, studying of Tesla's heritage is not related only with history of physics. We shouldn't look for technological aspects only; its essence is in miraculous human creativity in science.

TESLA'S SCIENTIFIC WAY NEEDS UNDERSTANDING AND FOLLOWING

We need analysis of Tesla's evidences relating with creativity lying in the base of new physical and technical processes discovered by him. It possibly leads to opening of new way of cognition. It is important to understand an original scientific and research method of Tesla and non-developed cosmic principles (relation), which are realized, with the World of ideas. Tesla could create visual images competitive with images really perceived by organs of vision. **He as if changes the usual direction of nervous impulse on the opposite direction, from brain to retina, eliminated external image and substituted it with eidetic one. Tesla sent his impulses like from "inside". He called it mental laboratory.** In addition, it was his main method of experimenting. He had an ability to transfer mathematical, abstract notions into inside visible images, give them geometrical interpretation, and than transfer them into the form that could be physically realized as working models for realization in device.

He "made corrections" and "adjusted" a device for operation in his mind. When later this device was made of wire or another material, it always worked. As Tesla said, **there was not a case** when such invention didn't correspond to Nature, i.e. it didn't work as a physical prototype.

His method was unique in comparison with other scientific prophecies. Michael Faraday, for example, like Albert Einstein felt a kinesthetic premonition in the moment of striking (it was pressure and motion in peritoneum region), he had some kind of mental stress in the moment when an idea appeared. Dmitry Mendeleev dreamed of periodic table in three dimensions colored with bright almost fiery red colors like it is in all prophetic dreams. However, it happens very rare, with few people and at certain moments. Tesla was in this condition all his life and in addition he has been training during decades to maintain his spiritual and creative activity.

Tesla was a man, whose main philosophy and axiomatic didn't belong to the modern world. We can attribute him to the epoch before Socrates, ancient philosophy. It was not an accident that he was born on Balkan Peninsula. Antique civilization appeared in the old days in southern part of this peninsula. Probably, the reason is different geomagnetic informational field, algorithm,

which is general to evolution of neural structures of Pythagoras, Plato, Zenon and Tesla.

Perception of Tesla's physics requires quite different understanding of mathematics; in some extent it should be sacral in the spirit of Pythagoras. Pythagoras considered that numbers and subjects are really interrelated. They correspond to each other in some properties due to informational and mathematical aspects of matter existence as one of manifestations of Divine Logos. Even less attentive researcher can immediately notice **that there are no infinitesimal or differentials in Tesla's works.** Tesla told about James Maxwell, creator of theoretical electromagnetism, that his elegant differential equations are the poetry. One could find Maxwell's works in Tesla's library; these works were published when Tesla was twenty and he was a student in Gratz. There is no another reason, which allows to explain, why Tesla didn't use them except one, very serious and deep: **he discovered more simple and more effective way and more felicitous physical interpretation of mathematical notions. By means of this notions he foretold course and results of experiment.** It is evident that he was a man of such strength and astuteness that he could keep away from conclusions of science, which was modern to him, its system of notions and mathematical apparatus, which were far from perfection. It is not occasional that he didn't uphold his diploma. It was an indication of spiritual independence for him, but not idleness or revolt as it can be sometimes.

His method of work could turn him to the absolutely different direction but it didn't happen. It proved that he could control the essence of his psychic visions and other conscious or unconscious manifestations that is the basis of creation. In the moment when he felt effects of impulses came from outside, he could override them, control and analyze. He could do an impossible thing: by means of exercises, which he developed himself, he got up to the level of man, who does not depend on impulses provided by external world, and at first on his own inner psychic states. Essence of his genius was in the fact that things understood by him and appeared in his subconsciousness under influence of external field was a picture of cosmic processes or the same natural laws. Because of it, he was distinctly different from other people.

He said about rotation of magnetic field, with which the rotor of an engine began to rotate under the influence of changes taken place in the external field: "Now I know, how Cosmos works".

This is the source of his theory on people as automates of cosmic forces. If we add Tesla's practically discovered axiom, that "energy of system is derived from environment", then we will find the reason of his conviction that entire Cosmos acts on the principle of vibration and resonance. Finally, if we agree with the fact that all **energy of system appears under the impact of external induction, than it will be easy to fancy Tesla's cosmological model as a series of circuits of**

magnetic fields, which rotate concentrically. Really, the Galaxy rotates, Solar system rotates around the center of the Galaxy, the Earth rotates around the Sun, and molecules, atoms and electrons rotate... All this is a range of rotating magnetic fields, which can be described by the only one law, the same law that was responsible for starting Tesla's inductive motor.

An open question is about **Tesla's real understanding of time and space categories.** We can say with certainty that in his research of how low frequency electromagnetic waves are transmitted through the Earth **Tesla didn't face with the problem of conjugacy of force and space, but, for example, general relativity theory (GRT) faced this problem.** GRT has some complexities in physical explanation of differential equations, which describe the properties of curved space under the influence of gravitational forces of different power. According to this theory, the constantly increasing attraction force can influence time itself returning events back and violating the principle of causality.

Rene Decart in creation of his coordinate system used the same geometrical notion to picture abscissa and ordinate. They are two straight lines interminable in their lengths, which he interpreted as time and space as if there is no difference between them. Is it possible? It is hardly possible. You know, space and time are different physical categories and we hadn't understood their nature enough. Besides, abscissa and ordinate are divided by an angle, which does not have a special physical interpretation. That's why if we are picturing physical phenomena in so geometrical way, we have a risk of not taking into consideration the criteria of reality in checking of mathematical hypotheses.

There is no doubt that Tesla by means of his physical research discovered a fundamental physical law, which is not clear yet, and applied it so easily and with a such astuteness, that a dilettante as well as a specialist could similarly understand that he spoke about exact knowledge. But now it is hard to imagine, what were these usual mathematical operations of division and multiplication, natural algorithm or square root taken separately, which are so significant in Tesla's experimental works. Physical explanation of mathematical methods, especially univocal, is situated on the boundary, from where we can see the field of transcendental mathematics. However, it is evident that **if all physical reality can be reduced to the mutual relation of electromagnetic fields, then mathematics is a theory, which can express the essence of these connections most effectively.**

If electromagnetic oscillations correspond to the mental plan (discovery made by Tesla at Colorado Springs is still proved experimentally now), **then it means that mental plan itself should have the same nature and follow to the resonance law.**

A conclusion arises that numbers are some "programs" of organization. Tesla's research, as it were, prove the verity of cognition theory by Plato.

In his theory, Plato stated that mathematics is a link between World of ideas and World of material phenomena. To be more precise, mathematical algorithms and mathematical logic is properly a method to realize ideas. Finally, all old innermost legends teach that matter is only a concentrated light, and it is Tesla's all-penetrating cosmic substance - "luminophore ether".

WARSHIPS AS TIME MACHINES

There is a story, book and two movies describing an event, when in 1943 American navy made an unusual experiment to make one of the warships invisible. It was necessary to make ship create a strong field around itself using power magnetic generators. This field should be able **to change the direction of light beams** as well as to grade the radiation from radar sets and thereby hide location of the ship. It took place as recently as six months after Tesla's death and disappearance of important scientific documents and apparatuses from his room. The event took place in Philadelphia seaport. When generators were engaged exceedingly, something unexpected happened. The number of strong electromagnetic fields **changed local time-space coordinates** and cruiser (Eldridge DE-173) disappeared for some time. Then the cruiser appeared for few seconds in another place, i.e. in one of the biggest sea base at Norfolk, in the port at south-east Virginia at Atlantic coast, which is 350 km far from Philadelphia. The strangest things happened to the crew. One half of sailors disappeared forever, some of them went crazy or got an ability to disappear and appear again by their own will. Part of the crew who survived, stated that they "have changed the world" and saw or even spoke to extraterrestrial beings. A thick greenish fog preceded to the disappearance of the ship. This fog covered body of the ship with starting of strong electromagnetic generators, which produced intensive focused electromagnetic fields.

Three ships took part in this experiment. The above-mentioned cruiser was in the middle, shell of bodies of other two ships served as a mirror. Actually, **the case in the point is open "lasers", which do not need vacuum. Beams of these lasers can be transferred at any distance and in any medium. Tesla produced such lasers in his New York laboratory even in XIX century. He illumined the room without any lamps by luminescence of air.**

Let's return to the significance of Tesla's experiments in Colorado Springs in 1899-1900 to see what are the real results of this work? When making experiments with ultra low and ultra high frequencies of electromagnetic waves, he obviously was successful **to determine the frequency and type of modulations, which is provided by fields of fine body of living people** as well as dead people (numerous articles are devoted to it, but these articles are lacking of theory and contain inexact suppositions). Most likely, he could create a field of ultra high frequencies, which corresponds to resonance frequencies of disincarnate

souls (it is soul, which already has no connection with the body) and thus he became proficient in visualization techniques of the so-called **astral level of being for biological organisms**. Calculations in Tesla's records could seem to be insignificant for modern specialist since they are very simple and cannot be understood without additional explanations. Only mind, which has similar perceptive ability, can understand Tesla's calculations.

SECRET OF "FIREBALLS"

Tesla could reproduce complex energy structures in laboratory conditions. He called the "fireballs". Russian academician Peter Leonidovich Kapiza also studied them, but he couldn't reproduce them in controlled mode without Tesla's resonance transformer. Nowadays physicists (Corum brothers in America) reproduce some of Tesla's experiments with a certain success. They could receive "fireballs" of a very short life period and only of three millimeters in diameter. Tesla produced "fireballs" of football ball size, kept them in hands, put them to the box and covered it with a lid, and then he took them from the box. These fireballs were almost stable structures, which could be kept for minutes. Certainly, Tesla knew much more about this phenomenon than modern science; he knew the secret of cold plasma fusion in free space.

Thus, Tesla penetrated to the field, which was unexplored before, i.e. the time engineering technologies. He also emphasized **that his electromagnetic waves were different from Hertz's waves**; i.e. the length of wave transmitted by him is equal to the distance on which this wave is transmitted. In other words, it is equal to the distance between sending object and receiving object. Hertz measures and space measures are changing in opposite directions: one Hertz is taken as corresponding to the wave length of approximately 300 000 km, 2 Hz – 150 000 km, 3 Hz - 100 000 km etc., which means that real length in space is decreasing with number of Hertz which are increasing.... That is why Tesla did not count in Hz-s... Obviously, Hz's measures are not adjusted to technical properties of electromagnetic tools.

*(Editor's notes: To my mind, it is not so important to produce calculations in Hz or in meters, but the difference here is the qualitative **difference between transverse and longitudinal waves**. In "Pioneer Radio Engineer Gives Views on Power", New York Herald Tribune, Sept. 11, 1932 N. Tesla wrote: "I have showed that universal medium is a gaseous body. Only longitudinal impulses can propagate to this body. They create variable compression and decompression like those, which are created by the sound waves in air. Thus, the wireless transmitter does not create Hertz's waves, which are a myth, but **it creates the sound waves in ether. The behavior of these waves looks like the behavior of sound waves in air** excluding the fact that the huge elasticity and very small density of this medium makes their speed equal to the speed of light." In other Tesla's article "The True Wireless" he wrote about this great*

*mistake of modern science: "**The Hertz wave theory of wireless transmission may be kept up for a while, but I do not hesitate to say that in a short time it will recognized as one of most remarkable and inexplicable aberrations of the scientific mind which has ever been recorded in history.**"*)

Besides, there were not only circles of oscillator (a device, which took part in the transference) in resonance, but the entire system was also in resonance with natural electromagnetic waves of the corridor of their passing.

It meant that **an oscillator, which started the transmitting, almost "intercepted ether" in the space between transmitter and receiver and by this way it created a field of standing waves. Thus, a waveguide formed at first, which couldn't transfer energy of its own. Then Tesla turned on a low-frequency field and transmitted waves of lower harmonic components of the main field carrier, in the ratio of 1:4.**

In such a way he could transmit energy at the desired distance and he could make strong continuous electromagnetic discharges in certain zones while creating the wall from ionoplasma. Nothing could fight through such energy wall without decay on molecules or atoms.

TESLA AS A CLAIRVOYANT

It is evident that Tesla knew the phenomenon that we can call parapsychology in the lack of a better word. As we told above, a method he used to make his discoveries or work in the laboratory has no analogues in the history of science. There are more that 150 000 documents in Nikola Tesla museum in Belgrade, but he didn't leave any description of his scientific method, which is comparable only with states experienced by yogis or with things known by saints. Now few people consider Tesla to be a philosopher or spiritual man. They do not think he was a man who inspired physics, technology and science. Finally, he founded a basis of a new millennium civilization with all his life and work. Though now his influence on the modern tendencies in science is minimal, his role needs in revaluation. Only the future will give a real explanation of Tesla as a phenomenon, because he went far ahead and he stands higher than the methods accepted nowadays.

A well-known Indian philosopher Vivekananda, on of the members of Ramakrishna mission, who was sent to the West to find out a possibility to join all existing religions, visited Tesla in his New York laboratory in 1906 and immediately sent a letter to his Indian colleague Alasingh. He rapturously described his meeting with Tesla: "This man is different from other western people. He showed his experiments with electricity. His attitude to electricity looks like an attitude to a living being. He speaks with it and gives orders to it. I speak about spiritual person of the highest degree. It is no doubt that he has a spirituality of the highest level and can call all our gods. In his electrical

multicolored lights, I saw all our gods: Vishnu, Shiva... and I felt the presence of Brahma himself".

PHYSICS OF TIME ACCORDING TO TESLA

Resonance is simultaneity, and if we change the periods of oscillations of electrical circuit, then geometrical picture of magnetic field distribution will change and generator is set in motion only under the influence of temporal factor. Motion is a result of asynchronism. It proves the fact that it is possible to obtain the change in speed of the motor and thereby its mass without an additional impulse, i.e. without an additional force. Tesla's reversible magnetic field can be a proof of this fact, isn't it?

Tesla wrote from Colorado Springs to his New York friend Johnson that he found a thought in "scribble" of high-frequency electromagnetic discharge and soon Johnson would be able to read his poetry personally to Homer, whereas Tesla would discuss his discoveries with Archimedes. The same way we can find a detailed description of "green" electromagnetic waves in his research journal. These waves looked like a fog, which appeared when the warship in Philadelphia disappeared. When Tesla returned to Colorado Springs, he told journalists that he made a connection with extraterrestrial civilizations. Only few people regarded seriously to this claim. However, there are evidences that Tesla continued his **research on "parallel worlds"** alone and silently without publicity of the results. To all appearances, he made a device to tune electromagnetic oscillations of his own brain, in other words, to control his mental activity. Thus, he could communicate with realities displaced in time (parallel worlds) without any problems.

These researches open a new page in the modern science. They show a possibility to merge material and spiritual aspects in human consciousness (matter and spirit in nature are already merged). **Tesla had an ability to develop both directions of uniform science. He studied physical basis of mind and mental basis of physics with the same success. He came to the essential general conclusions that matter consists of organized parts, which are manifestations of ether in its excited state. One of these manifestations is electromagnetic oscillation.** That's why resonance law is general natural law and it's of time nature, too. A connection between phenomena is provided by resonance of different kind. Electromagnetism is the basis for these resonances. Needless to say that from the one hand, if physical reality can be reduced to the interrelation of electromagnetic fields, then mathematics is a theoretical expression for these relations. From the other hand, if electromagnetic oscillations interact with mental plan, which has the same nature, then numbers are some reflections of organizational structure of electromagnetic fields.

(Editor's note: We discussed the above-mentioned possibility to create "energy bridge" between different levels of reality, i.e. between two local spaces of different

speed of time. Now we should to clarify one more interesting Tesla's idea – "parallel worlds". Possibility of co-existence for two or more Universes in the same place can be physically and mathematically proved. The "time shift" is a new notion, which should be used for it. There is an old Shrodinger's conception of "flickering matter", which describes reality and allows superposing of two or more material objects in the same place with some time shift. The matter in this conception dynamically appears and vanishes. This principle mathematically is very close to theory of modern multi-channel telecommunication systems with time separation of signals. Period of time when the matter is "appeared" can be only small part of the cycle and by this way many "parallel Worlds" can be organized. The time course or rate of processes in all parallel Worlds should be the same but they should be asynchronously organized. So, this aspect is different in principle from idea of "energy gradient" between slowed space-time and accelerated space-time.)

I personally believe that mathematical and physical truisms, or rather geometrical and arithmetical phenomena, should join and become a basis of uniform Cosmology. Now we should take into consideration that we are staying on the threshold of this Cosmology. Somebody calls it a "New Science", but "Unified Science" is more appropriate term, because unified laws make the basis of our Cosmos. In fact, Unified Science is the Science of Time.

Tesla's electromagnetic theory was vivid illustration of unification of material and spiritual origins of the Universe. Practically he made a lot and due to this fact he did not leave his theory for us. Possibly, he could leave us a religion but he didn't want to do it since he knew that God needs conscientious followers in science.

"MYSTERIOUS PILGRIM" BY MARK TWAIN

A famous writer Samuel Clemens (we know him under the pseudonym Mark Twain) often visited Tesla's laboratory. Tesla was very close to him and several years after Twain's death he spoke about him as if he was alive. The writer Mark Twain died in 1910, and his enigmatic novel "Mysterious pilgrim" was published six years later.

In this novel, he told about an Angel who left heavens and came to a small Austrian village. There he met a group of boys and let them into the secret of the universe. If we recognize little Austrian village as Smilyany and figure of Angel as Tesla, then an explanation of a very strange theory on human destiny and cosmology will appear. This theory expressed in this short story made it to be different from other stories of this writer. Angel tells about the source of all people's troubles and misfortunes. **All it comes from misunderstanding of true sense of occasionally very small events.** Each of these events defines future links of the next events. As he considered, people's free will is an illusion since all is predetermined and will come to a principally predictable result. That's why the given

human individual psychosomatic structure has limited number of possible destinies (ranks of events) and a man can switch back and forth due to the will of higher beings. For example, Angel takes part in the event, when a person closes or opens a "window", which was predetermined to the person, so after this action the destiny of this person will go another way depending on the specific action, and this act will call a new succession of events. This entirely coincides with Tesla's idea of a man as an "automaton of cosmic forces" and it is brought out clearly by simple dramaturgic tools used by famous writer.

Finally, before Angel left his friends, he let them to the last magic secret, which would be terrible to them, i.e. secret of Non-existence. He said that all is only a thought. There is nothing existent... I am only a thought, lonely thought, which travels along the empty space of the Universe".

MYSTERY OF WARDENCLYFFE TOWER

Let's visit Tesla's laboratory in Colorado Springs again, where inventor made his experiments using electromagnetic field of high frequency. He studied operation of human brain on himself. He wrote to his fellow Johnson about some intelligent properties of electromagnetic field. **Biophysical principles of neurology were born.**

Just after this, in 1901, a construction on Long Island began. The matter concerns a complex transmitter of electromagnetic waves. Tesla applied all his knowledge to design it. They are: teleautomatic control, Hertz's wireless wave transmission, **unique ether theory, according to which any natural element of Mendeleev's system has its own gravitational acceleration and Tesla's techniques to control the speed of time course.** A new physics was born.

A question appears what concept of the World lies in the basis of experiment with resonance focusing of magnetic and electromagnetic fields. Let's return for a moment to a mystic Monard and his picture of Cosmos. Speaking about vibrations, he underlines the presence of different nations, races and tribes on the planet, but we cannot meet people, who could live without music and dancing.

Even 2500 years ago Pythagoras said to his progeny that **stone is a "frozen music"**. Every particle of the physical Universe has a certain property, which is pitch and pitches of its separate frequencies, i.e. its melody.

Nature of influence of music on a man is the same as the nature of its influence on the world of things and events, because vibration is the basis for all. Frequency of vibrations of subatomic particles is very high and waves in the center of subatomic nucleus move quicker. Vibrations of all radiant energies (radio waves, heat, light, X-waves and so on) could be ordered. Electromagnetic spectrum in such consideration

contains more than 80 known octaves and visual light is only a part of this rank.

Electromagnetic waves as acoustic waves in music have their own harmonious high-pitch tones and there is some kind of octave principle, i.e. doubling of frequency. Laws of music and harmony applied to electromagnetism give splendid results. That's why Nikola Tesla used the works by Herman Helmholtz about sound to develop his electrical resonators.

Johan Kepler, an astronomer, who lived in XVII century, believed that every planet of Solar system is alive and there is a guardian angel on every planet, which listens its music. Planets "play music" on their circumsolar orbits.

(Editor's note: Dr. Kirill P Butusov, St.Petersburg reported about his discovery of correlation laws for all planet orbits of Solar system. They really confirm the laws of harmony.)

A particular combination of sounds, which we call music, is an expression of relation between things, in other words, their likeness and difference. It is also a principle of appearance and disappearance of phenomena as well as the principle, which shows how new young systems grow from coordinated harmonious systems. The deep inner structure of music is the same as the structure of all the rest.

A Dutch scientist Christian Huyigense in 1665 noticed that pendulums of two clocks hang on the wall near each other starts to work in rhythm. It is a universal phenomenon. When two or more oscillators begin to pulse with sufficiently small difference in time (with small shift in phases) their oscillations come to coincidence spontaneously. They behave according to the principle of minimum energy since every separate pendulum requires less quantity of energy with synchronous pulsation than in the case of arrhythmia. This co-ordination is everywhere, but we rarely notice it. We can say that all animate things are oscillators, which pulse and change their rhythms. Even the simplest unicellular organism is in complex oscillatory state. All motions on subatomic, atomic, molecular, subcellular and cellular levels are coordinated. In such an organism like a human one it is very difficult and even impossible to determine corresponding parameters. Our inner rhythms are closely interconnected and they are coordinated with external world. Physics of a man and states on his rarified plan (astral) change in one rhythm with the Earth's movement around the Sun, with ebbs and flows, change of day and night and with many other cosmic rhythms. With violation of coordination between these rhythms organism feels discomfort and even presentiment of near illness.

Our physical body serves as an exponent of our inner leading pulse and at the same time, it is our individual distinction in Cosmos. **The so-called "fine body" in**

essence has an electromagnetic material base. Life of each of us is characterized by certain rhythmical scheme, which is subjected to physical death and retains an extra-body experience of the spirit, which should be reincarnated.

There is a supposition that the so-called "Shuman waves" takes place in the telepathic transmission of image and thought. These are waves of 7.8 Hz frequency, which form a field of standing waves in the space between ionosphere of the Earth and its surface.

(Editor's note: There are two bands of the Shuman waves, i.e. short waves Earth-ionosphere and long 7.8 Hz waves around the globe. There is a conception, which considers them as different sides of the same phenomenon: the unified helical spiral wave in spherical resonator of the planet. In this case the short waves correspond to diameter of the spire and the long waves correspond to radius of the resonator. Why it is important? In biology we can see many examples of helical structures, for example, the DNA molecule. Shuman resonance can be considered by analogy with biology and processes in the global resonator of our planet become a kind of vital processes.)

It may occur that a brain enters the state of resonance with some external structure; a result will be appearance of its image at a distance, i.e. there is a contact made by means of resonance but not by means of radiation. Since these waves have a big length (38 000 km), then transmission takes place immediately. Waves with such an extreme low frequency cannot be screened by usual means. In essence, waves of an organism often coincide with these waves by phase. However, the problem is that waves with such low frequencies can transmit quite little information and therefore it is often impossible to receive a clear picture and give a full description of images of things.

(Editor's note: In this point I cannot agree with the author since the low frequency waves can "transmit quite little information" only for Hertz's case of standard modulation methods. Another case is longitudinal waves. For this case the modulation means changes of density of energy in point of receiver and it is not related with the wavelength. The analogy for this case is changes of electric field. If intensity of electric field is modulated by some technical way, then an electroscope will feel this changes as instant signal at the moment in any point of the field. In this case the wavelength is zero but frequency of changes of the field can be a very high frequency. Different information can be received in nodes or in antinodes of the standing wave and it is the reason why some places of the planet are better or worse in this aspect. In some articles it was discussed as scalar field modulation.)

Modern holistic (integral) approach to understand the Universe has sacral traditions per se. These traditions were laid in foundation of any religion, which are Judaism, Islam, Buddhism, Christianity, where we can find a clear **statement that every part of Cosmos**

includes the whole. It is analogous to the principle of holography. It is considered that we can make a full hologram of the Universe based on any of its small parts. Formation of this hologram can take an arbitrary long time. Hologram of a **real** man is something different from physical basis of **ideal** cosmic rhythm.

(Editor's note: Yes, the more close to the ideal – the less the difference.)

Consciousness, which is a continual manifestation of rarified substance (astral) of a man, is shifted by its phase in comparison with genetic material, and conscious will is only a one of universal instruments of cosmic purposefulness. A "right" man acting in a "right place" is a man whose intentions are coordinated with cosmic rhythm. Conscious intention has a big power, and our main choice is to take a responsibility for its results. So, **actually we are made of waves but not of matter, which is called "solid". All that we call things and events are a manifestation of acting of these waves.** Every man has its unique cosmic identifier assigned by a certain function of waves, which represent us. Every man has his own hologram, which contains universal information about time. Identifier is a main element of Cosmos that characterizes an individual specificity of a person, who using this identifier. By means of this element the person can influence on everything by methods, which even seem to be unusual. We are a peculiar hologram, so potentially we know all about all.

After above-mentioned common statements, let's remember about Nikola Tesla. Secret of Wardencllyffe is deep-hidden, and it is impossible to speak about its real destination now. Nevertheless, many things show that the object built by Tesla in Long Island (that time it was a New York suburb) went ahead of all dreams of modern scientists. It was a translator of electromagnetic wave system, which could be easily checked. By means of these waves that scientist called earthquakes, "mental" shifts in people and animals, fired the atmosphere, made impenetrable energy barriers in ionosphere, controlled time, split and condensed the clouds by means of corresponding frequency of waves and, finally, he got an inexhaustible energy from ether using a principle of "ether" technology, which is still unknown for us. Great Master Tesla demonstrated his experiments to numerous witnesses, but he never announced all results and devoted anybody to his scientific principles. Real reasons to build this giant electromagnetic and mechanical oscillator were the motives much more serious than we can imagine.

A report by W. Pickering, professor of Harvard observatory, named as "Alternating light from Mars", was published in "New York Times" on January 16, 1901. It may be shortened like: "Last year in the beginning of December we got a telegram from Lowell observatory at Arizona. This telegram informed that a strong light flux was noticed coming from Mars and lasted for 70 minutes. Lowell observatory specializes on observations of Mars, and observer was an attentive,

approved and experienced man, so we have no doubt in his statements. We will consider these facts immediately and will notify Europe and America via telegraph. The light came from the known point on Mars and there is no unusual in this point according to scientific data. Whatever there was, now we have no means to research anything. We cannot say with certainty if it is a sign that there is an intellect. For the time present this phenomenon is inexplicable.”

When this information appeared, Tesla was in New York. He just returned from Colorado Springs, where he in the course of year made epochal and inexplicable experiments with electromagnetic field of strange resonance property. (Tesla’s laboratory in Colorado was situated precisely on the summit of Pike’s Peak. It is interesting that Indians Hopi deified this peak, they consider it a Spiritual Pole of the World).

Encouraged by the information he got from Lowell observatory, Tesla stated in the gazette that during his research in Colorado Springs he communicated with Mars himself. That time he made hints about that he already designed a device available for interplanetary communication, which will be improved in future. He wrote: “I’ll never forget the first feeling that I felt when I understood that I have a contact with something, which will have priceless and immense results for all humankind. Thing that I noticed, frightened me as if I saw something mysterious and almost preternatural. Little by little I understood that I am the first who hear messages sending from one planet to another...”

A respected Professor Holden didn’t consider Tesla’s contacts with extraterrestrial beings by means of electromagnetism, because he thought it was not a serious case. Tesla did worked with some extremely long electromagnetic waves of a very high frequency that does not correspond to the accepted theory, because increase of wave frequency is always related only with short-wave transmission. However, Tesla’s waves were different from those discovered by Hertz. They propagated with a speed, which far exceed the speed of light, better to say that the propagation is instantaneous.

(Editor’s note: We have already discussed this principal difference in modulation methods for Hertz’s waves and for longitudinal waves. In this case the SETI program is not ready to receive any signal if their telecommunication specialists use Hertz’s waves. It is necessary to develop longitudinal waves telecommunication equipment for SETI and other projects.)

Tesla has his own theory of electromagnetism, which still is not interpreted. He discovered not only the possibility of wireless transmission of energy through the Earth and atmosphere without any losses, but also he proved an unprecedented “breakdown power” of these waves in spreading through space. There is an unconfirmed legend that Tesla was the first who sent periodical signals to the stars. There were coded geometrical theorems like theorems by Tales,

Pythagoras, and an Archimedean formula of harmonic series (we speak about addition of the series $1 + 1/2 + 1/4 + 1/8 + \dots$. Sum of this series is equal to two but number of terms can be infinite). Three days later Tesla was surprised to get a response. He guessed the principle, which lied in the basis of response code, and got a human face with regular features. At first he couldn’t understand, if those who made this picture and sent this signal demonstrated that they know our civilization, or they drew their own image to demonstrate that there are similar beings in Cosmos. If it is true, then it should be clear that Tesla refused from any open discussion of this discovery because he couldn’t get a due reaction. But soon something occurred that returned discussion of this problem in American press.

In 1902, a famous English physicist Lord Kelvin visited America. When spoke to Tesla, he expressed a full consent and understanding of the fact that details of Tesla’s “extraterrestrial communication” were not for public discussion. After the banquet organized in his honor, Lord Kelvin said some good words about New York: “It is the city with the best illumination in the world and it is the only place on the Earth, which can be seen from Mars”. In addition, as if in some kind of inspiration he finally exclaimed “Mars is signaling New York”. This time nobody had objections. Even persistent Professor Holden kept silence. It was not polite to contradict to such an old and respected man as Lord Kelvin. Instead of objections an article written by the writer and publicist Julian Hotorn, Tesla’s associate, appeared as a direct conclusion from this discussion. He described Tesla’s sensational methods as a fantasy. Hotorn wrote: “It is evident that people from Mars and other old planets had been visiting Earth for years and attentively observe the development of our civilization. Besides, Tesla’s birth changed all. May be, it is possible that people from the stars control his spiritual and scientific development, isn’t it? Is there anybody who can know something concrete?”

In the heat of the disputes concerning Martians Tesla was alone in his Tower and continued to work hard and insensibly on the biggest project of his life, Wardenclyffe Tower to make radical changes in development of our civilization. These changes cannot easily be considered earnestly.

EARTH PLANET AS A RESONANCE SYSTEM

When made a decision to build a translation station, Tesla at first bought a plot at Long Island and then asked a famous manufacturer Morgan for help. Morgan was a member of numerous secret organizations; also, he was a genius of business organization. He already had cancer and had a hope that Tesla’s science would help him to recover. He expected that Tesla somehow would connect him to some vital energy sources and would make him young and healthy forever. Tesla thought that Morgan should be a man who would maintain this plan. He considered that a man has predestination of his life collisions and every man is predisposed to its own activity.

Negotiations between two old friends were private and hard. There is no clear information about the time when Tesla decided to begin the construction of Tower and about expectations of Morgan as an experienced businessman.

During his public appearances, Tesla changed his statements on destination of the Tower at Long Island twice. At first, he stated that the question is about universal telegraph and telephone system of wireless energy transmission through the Earth. Technical properties of Wardencllyffe do not prove neither of them. The truth could be found in his letter to Morgan: „What I contemplate and what I can certainly accomplish, Mr. Morgan, is not a simple transmission of messages without wires to great distances but it is the **transformation of the entire Globe into a sentient being**, as is were, which can feel in all its parts and through which thought may be flashed as through the brain...”

Many authors, especially Tesla's biographers, blamed Morgan for reduction of financial assistance in the moment when Tesla was already on the threshold of very important discoveries, when it was necessary to finish and start his masterpiece, Wardencllyffe Tower. But Tesla in his autobiography clearly tells us about it. In spite of all that was said, Morgan fulfilled all his obligations regarding him. “My project was retarded by influence of natural laws. The World was not ready to accept it. It was too far ahead of time. But the same natural laws will prevail in the end and the project will be repeated with a triumphal success.”

Tesla tested his transmitter at its full capacity on June 15, 1903. He began his experiment exactly at midnight. This night New York citizens witnessed the event, which was outstanding for future science. Glaringly bright strands of electrical plasma with the length of more than hundreds miles connected the spherical cupola of Wardencllyffe with the sky. The next day “New York Sun” wrote that people living near Tesla's laboratory at Long Island were interested very much in his experiments with wireless energy transmission. Last nights they were the witnesses of very strange phenomena, multicolored lightning made by Tesla himself, then inflammation of atmosphere layers at different altitudes and along the big territory: “Night suddenly turned to day. Sometimes air was full of luminescence concentrated along the edges of human body, and all people radiated a mysterious shine. They seemed to be a ghosts.”

According to the initial Tesla's plan, five towers, the same as Wardencllyffe, should be built. The second tower should be constructed in Amsterdam, the third in China and the fourth and fifth should be built in the North and South Poles. However, this plan was postponed due to the reasons that should be investigated. Finally, if all this could be realized, then what would be achieved? **The Earth would become a uniform homogeneous system, which could be controlled by telephone commands destined for**

starting of certain oscillators, which produce and transmit electromagnetic waves of various frequencies. The same electromagnetic system should be completed with a system for mechanical resonance with the Earth. These systems consisted of a tunnel passed under the laboratory and filled with water and oil, and hydraulic pumps served as oscillators. By synchronized actions, based on exact mathematical calculations both systems could start vibration simultaneously together with rarified stratosphere, ionosphere and atmosphere as well as with liquid and solid structures of our planet.

Before conjecturing relating to all possible consequences of such event, we should give two more examples that are unusual. In his interview of July 17, 1930 given to “New York Sun” Tesla redoubled his enigma stating that “people who live near Wardencllyffe and were frightened with my experiments could see much more...if during last two years they were more awake than asleep they could get acquainted with really incredible things. Sometimes, but not now, I'll announce something, which you cannot find even in fairytales.”

After unusual night, when he fired the sky above not only New York but also above all vast area of Atlantic Ocean during his experiment, Tesla suddenly left his laboratory without a clear reason and left all things untouched. As we know, he never crossed the threshold of Wardencllyffe, never visited it and never appeared on this region. It was very strange, he didn't take out of the laboratory any calculation, drawing or document, none paper. It was a turning point in his public scientific work. He was living for 40 years continuously working, but he patented his mechanical discoveries only and published only the articles in newspapers.

TESLA'S ETHER TECHNOLOGY

Tesla discovered and used the law, which concerned fundamental properties of Ether, and the law, which structure the initially infinite and homogeneous ether. Supposition made about continuity of ether as one of the main cosmic mediums means that **imaginary “center” of Cosmos is everywhere, that law of structuring of such medium should have analogues with the law of location of points at geometric spheres.**

(Editor's note: Let's take a sphere... Two points of diameter, three peer point of equilateral triangle, four points of tetrahedron and so on...)

There is no information that Tesla formed and told his theory in such a way, but he left some systems working on the principles, which are unknown to modern physics. The question is about **ether generator, i.e. electromagnetic ball, which has constant electrical potential in spite of energy leakage**, then about a **synchronous motor working on gravitational waves of the planets of Solar system** (this motor turns on independently at the certain season and reacts on the corresponding arrangement of planets. It also

independently turns off, when resonance interval of time ends). Besides, Tesla designed a **metal disk, which could levitate at some height and has antigravitational properties**, which do not depend on gravity force of the given place.

Parts of the structured ether can be in resonance or not. In the first case a condensation of subatomic particles, such as electrons, protons, neutrons, takes place. **These particles appear by means of resonance synthesis of photons, according to the same principles as during formation of photons as specific particles of ether. Asynchronous, non-resonant parts of Ether form a space, in which resonating photons form matter.**

(Editor's note: This is an important point, which related with notion about parallel Worlds. Each World is a totality of elementary particles, which appeared by means of resonance processes from other non-resonant part of Ether. Different frequency means different type of matter. But elements of the same matter can co-exist in the same space "in parallel" with some time shift between oscillation processes.)

Tesla's Wardencllyffe Tower was a "Pythagorean" oscillator. Mathematical description of transmission of particular electromagnetic waves was identical with Pythagorean creative method. Signs used by Tesla in his equations have a single-meaning physical interpretation. Principles of Tesla's ether technologies belong to the level of cosmic existence, where it is possible to control space and time. **The principle of resonance and harmonic oscillation of ether seems to be so clear that all problems of modern physics, especially problems of energy conversion, will be solved with its development.**

By means of his vacuum tube **Tesla got protons, electrons and neutrons directly from physical continuum (ether) and reproduced them at any distance.** Instead of giving a possibility to the bundle of protons to move free through space to some place, he created conditions for momentary appearance of arbitrary quantity of particles in the given place. **Quantity of protons, neutrons and electrons was unlimited and difference in their quantity was stipulated by created shift in time.**

(Editor's note: Modern physics know how it is possible to create mass particles, for example, electron-positron pair, from physical vacuum, but they can not organize it without huge energy source of the beam accelerator. Tesla's resonance method is a real way to unlimited free energy sources by means of structuring of the vacuum. The first step to this technology is development of the theory and experiments with the longitudinal electromagnetic waves. Really, the theory of acoustic resonators already is well known. All we need is its application to electromagnetism.)

Starting from suppositions that Tesla knew some unusual for science and still inconceivable laws of nature, we should think about what technical goals he

followed using Wardencllyffe Tower and what consequences he expected?

1. To call oscillations of ionosphere. Since the difference of voltage between surface of the Earth and ionosphere constitutes about two billion volts, then the Tower is continually oscillating on upper and lower harmonics of oscillations of ionosphere up to full coincidence with it by phase (up to resonance) and can discharge it in the same way as usual electrical condensers are discharged. Due to the reason that such a charge could pass immediately, there would be a real threat to New York. A high-energy plasma column of a very big diameter (possibly of several hundred kilometers) would appear from ionosphere, and all situated in this space would be disintegrated. Of course, Tesla had no intention to destroy New York. He only wanted to take energy from ionosphere by means of short oscillatory intervals to recharge his huge oscillator, which required maximum voltage of one hundred million volts. As a system for hyperfine oscillating, Tesla used **a huge quantity of ultraviolet lamps, placed at the top of his Tower.**

2. To take energy from Ether. It is possible to shift time coordinates and produce an energy transmission by time cycles from the Past to the Future without any violence of electromagnetic equilibrium of environment by means of oscillation of non-resonance electromagnetic fields.

(Editor's note: Usually the term "energy transmission" means some distance in space between two points. In this case it is a case of two points in time, i.e. some moment in Past and some moment in Future. So, we can use gradient of energy in time (chronal gradient) instead of spatial gradient (voltage). I discussed this method in my early publications and some technologies were reported during St.Petersburg's congress New Ideas in Natural Sciences, June 1996.

Of course, it should be done with strict following of mathematical law, which determines the interrelations of frequencies with a great accuracy. In an opposite case, the results can get out of control and an unexpected destruction of material and biological objects or their momentary drop out from segments of our reality can take place like in Philadelphia.

3. To open temporary windows in "parallel worlds". Since parallel worlds have an electromagnetic structure with the wavelength and frequency of oscillations different from those of the Earth, then it could be possible to make some connection between frequencies of oscillations of our world and other worlds by means of complex harmonic oscillating. Due to this, some pictures of these worlds could become apparent in our earth conditions (and vice versa).

4. **To accelerate human evolution.** If we make a permanent high-frequency field harmonizing with collective electromagnetic field of people, then little by little it will be possible to achieve an increase in sensibility of perception and increase in ability to receive ideas. But radiation of these frequencies could be harmful since any disharmony of this field (non-resonance states) calls a partial separation of astral and physical substance of a man, it can lead to sickness (for example, cancer) or mental disorders.

Full separation of one substance from another, of course, leads to death. In addition, all these processes act slowly. Due to their lack of knowledge, people pollute their electromagnetic environment more and more by radio waves, microwave radiation from locators (which call cancer, it was proved experimentally). Other electromagnetic radiation also has negative impact if these electromagnetic waves are in disharmony with frequencies of oscillations, which are natural for vital functions and for informational field of Solar system.

5. **To be able to transfer the Earth planet into "parallel realities".** If Tesla could finish the construction of system made of five transmitters and start it, then he could make the whole Earth to be in the state of oscillating as a uniform resonance structure, and such oscillation, which is peculiar to some other reality, and thus physically move us to the "parallel world". Thus possibly, he wanted to accelerate evolution of civilization. However, it could be very dangerous, because if the Earth enters the state of resonance oscillating with "parallel worlds" and it takes a long time, then catastrophes could occur like that was described in the myth about Atlantis.

Modern experimenters should keep in mind that secret of physical world lies in metaphysics and as long as our science will be blinded with research of "crude" matter only, we will be in the state of illusionary split consciousness.

It is necessary to provide a breakdown of human consciousness, and it will happen in physics. We should free human mind and adjust it to deep cosmological research.

Ethics is cosmic principle, which contribute to energy distribution, isn't it? Then it can get a status of natural law. And we will be able to explain to the supposed habitants of worlds, which are shifted in time, why we can influence on the Galaxy and stars by means of special reactors and why we take part in experiments, which directly change us.

If essence of ethics is a harmony and kindness is state of energy equilibrium, then mathematical, cosmic ethics certainly acts in the world. It is impossible to violate mathematical laws. "There is no czar's way in geometry" – said Euclid to the czar Edip, who

worked on solution of difficult geometrical problem. Cosmos itself seems to be a big experimenter. Our mind asks it the questions, which are clever or not. That's why every true philosopher and scientist should reject the applied science appeared from defect and rigid theory, when he studies something concerned with cosmic phenomena.

Evolution of Tesla's philosophical positions beginning from oracular one and finishing with engineering ones, beginning from engineering ones and finishing with metaphysical ones, as Tesla's private life, these all are the milestones on the way of formation of a new paradigm of scientific and technical civilization. His attitude to people and himself is a result of transpersonal absolute manifested in him. He had no personal attitude to anybody including himself and that's why he did little mistakes during his life.

Tesla is often cited in modern science. The scientists, who didn't notice contradictions of Maxwell's electromagnetic theory until now, began to study Tesla, especially those who work on unification of electromagnetism, gravitation and strong and weak interactions. Is it true that matter is infinitely divisible or not? If space could be divided infinitely? Finally, what is the role of time in physical processes? May be time is only a measure, usual coordinate, as it is supposed in modern physical theories basing on the theory of relativity and quantum mechanics.

(Editor's note: Dispute about "substance time" or "mathematical time" can be solved in Ether Theory if density or direction of ether define parameters of existence of any matter, i.e. its time. In this case we need to discuss the time itself no more. We have to discuss ether and its local properties.)

Tesla's ideas about ether as a basic medium, which structurally joins space and matter, were rejected and it led to stagnation in physical thinking and to continual returning to out of date ideas. In scientific circles, which take a risk of creation of a "new physics" Tesla's concepts now become increasingly valuable and they are studied. There are experiments with ultra low and ultra high frequencies of oscillations, which have specific modulation and form the radiation of ionosphere. All possible consequences of it are still unknown. This includes control of meteorological phenomena, especially hurricanes and atmospheric pressure, **infusion of thoughts at a distance by means of electromagnetism, i.e. resonance with electromagnetic transmitters.**

It becomes evident that it is technologically possible to imitate external electromagnetic manifestations of human thinking. It is the position of Tesla's cybernetics. "A man is an "automatic machine" of cosmic forces" – underlined Tesla in his main articles and lectures. Experimenters can say that it is much easier to excite mass emotions of people by means of corresponding oscillation of ionosphere, which includes harmonics of collective unconscious state of

all humankind. The ionosphere is a key to control mass feelings and thoughts. Tesla understood all this yet in far 1899 living in Colorado.

Modern university science and in particular Serbian scientific community reject Tesla's ideas mainly because it cannot understand them. The second reason is the opposition, which Tesla demonstrated to nuclear physics, and his prediction that application of this technology would not be continuous and quantum mechanics has no serious scientific future.

Tesla as cosmologist had his own philosophical and religious position: Aristotle stated that there is an independent higher spirit in cosmic space - entelechia. This spirit sets matter in motion and a thought is his main attribute. In the same way, Tesla believed that uniform Cosmos is unified in material and spiritual sense. There is some kernel (core or center) in cosmic space, from which we take all strength, inspiration, which attracts us forever. Tesla felt its power and its values, which it sends all around the Universe and keeps it in harmony. He couldn't guess the secret of this kernel but he knew that it exists, and when he wanted to give it some material attribute, he ascribed that it is LIGHT, and when he tried to understand its spiritual principle, then it were BEAUTY and SYMPATHY: "A man who keeps this religion in himself, feels strong, works with joy because he feels himself as a part of universal harmony."

RESUME ON PERSPECTIVES OF TESLA'S SCIENCE

Not only physicists, who continue Tesla's work, but all people who think a lot about scientific problems, are agree that modern physics is really a contradictory system of knowledge. From the one hand, time is relative and depends on the observer. From the other hand, it is stated that time of quantum transfer cannot be measured. Nuclear reactions are studied on cathedras of history of physics. In the course of these reactions transformation of matter into energy goes without a residue. In astrophysics, full transformation of matter into energy is not possible. It is known that mass calculated as an effect of the gravity force differs from the mass calculated from the speed of rotation of the stars around the center of Galaxy (we can get it by means of "Doppler's" shift of lines in spectrums of their radiation). Therefore a significant defect of visual matter or mass can be found, and it amounts to 90%. Instead of changing the theory, scientific schools speak about "defects of mass".

Tesla worked on special "vacuum tube with an open tip", which serves to transmit particles at any arbitrary large distances. The targeting was made by electrostatic way by the lines of electromagnetic layers of atmosphere, which can be distinguished by the naked eye. It is the layer, where clouds float.

*(Editor's note: We have discussed above **it was not a "transmission"** in classical sense but it was the **resonance structuring of physical vacuum** in some*

distant place and any particles can be formed by this way.)

It became known that multiphase system, inductive motor and generators of alternating current are the less significant Tesla's discoveries. **The most important should be those, which relate to his experiments with ether and time.** We can underline a perfectly new cosmological physics, which is based on electromagnetic resonance of ether, on resonance effects of time.

Main Tesla's discoveries are the following. The first is **transmission of energy** at any distances. Really, it is a superconductivity of **natural environments**. The second one is **"fireballs" (synthesis of structural elements of ether and matter)**, high-frequency resonance oscillator adjusted for waves, which are different from Hertz's waves, **waves with the so-called longitudinal modulation**. It is a kind of free choice of consciously selecting electromagnetic units, which initially and elementary codify working electromagnetic field of human brain, modulate brain oscillations or change their nature, and in sensitive aspect it leads to various emotions, changes in consciousness, creative impulses, super perception up to super cognition. The third one is **dynamic nature of attraction**, in which every element of periodic table has its own constant of attraction; in this case Tesla continued the works by Etvash and derived several universal methods. And finally, **Tesla's electromagnetic theory**. This theory was not explained or published. This theory does not use the generally accepted notions, such as "energy", "wavelength", and "frequency". Instead of it this theory introduces the notions of "distribution curve", "vibration of spiral systems", "electrical pressure", "proportion of transmission", "ether", "dynamics of electromagnetic fluid", "geometrical abilities of tube" and so on.

From modern mathematical arsenal Tesla used only Fourier's ranges to separate electromagnetic oscillations into higher and lower harmonics, since it matched his main supposition on simultaneity of action of all parts of infinite ether. It is clear that infinity cannot be formed with nonsimultaneous parts.

Tesla's cosmology is a cosmology of a general type, which relates to isotropic and infinite Universe. At the same time, Maxwell's electromagnetism can be applied only at comparably small distances. In general, Maxwell just made a mathematization of Faraday's experiments, but he didn't build a final concept of this problem. Due to the incompleteness of Maxwell's theory, great difficulties in Special Relativity Theory appeared (interrelation of speed of propagation of energy and speed of light). Einstein himself ignored relativist changes of space-time with motion in photoeffect, since if energy changes together with speed, then there are no effect, if energy equivalent of electron trajectory (knocked out by photon) is absent. In general theory of relativity shortcomings of Maxwell's theory led to an idea of "world line" (in essence, it is a pancosmic extrapolation of magnetic force lines, which are

visualized by iron filings on laboratory tables) and, finally, to the notion of Cosmos finiteness, which alternately shrinks up to the "point", which is located out of dimensionality, or explodes... unlikely development.

It could be surprising not only for Tesla, but also for many antique philosophers and scientists, Pythagoras, Plato, Euclid and others.

Nikola Tesla and his science are very significant for the modern world of telecommunicational systems and as a way out of crisis of scientific and technical civilization, which needs philosophical ontology. During hundreds and thousands years, we had many great scientists, but nobody has a temple built in his honor. **Such temple is built in California, in San Diego, in the one of centers of world scientific and technical power.** Neither Plank, Einstein, Bor, Pauli, Torichelli nor Newton in spite of their interest in alchemy, theology and mystics, none of them could convince people of transcendental significance of their scientific views.

Christian civilization is based on suffering, sacrifice and atonement of Christ, on the mystery of christening, resurrection and the afterlife world. It is full of human feelings and emotions. Technical epoch under the badge of Tesla is a humane symbol of spirit and matter unity and it establishes not a religious belief, but belief in scientific principles, and gives a discernment in fundamental laws of Cosmos. **Historically, Tesla radically changes the form of religious belief on the planetary level.** That's why he is not only a scientist but also a universal and historical phenomenon of a higher level. **He changed spirituality of theology to the spirituality of science. Besides, modern physics already became a philosophy of new times, since it openly works with the structure of matter, genesis of Cosmos and, recently, Ethics.**

Quantum mechanics of Copenhagen circle including Bor, Heisenberg and Pauli does not give an answer on the question about structure of main unit of matter. Theory of relativity doesn't solve the problem of time substantiality and doesn't explain the essence of force concept. Tesla's physics relating to the real physical space or ether, and his scientific thought derived from positions of antique cosmology and Pythagorean mathematics (not from electromagnetism of XX century) couldn't be continued as long as appeared other limits on the way of creation of general physical theory, which includes events of physical reality. After Chernobyl all people understood that nuclear reaction is nothing more than irresponsibility without real understanding of matter structure. And tragedy of spaceship Challenger reminds us that humankind still does not have enough experience in long space travels and jet engines cannot solve the problem. It is better not to speak about medical aspects of interstellar travels. Tesla's physics gives answers and instruments to use it in cosmology.

Sometimes a notion that we use to denote some physical phenomenon didn't exist in times of Tesla. The same is with the method of introduction of mathematical

symbols to make calculations. All it gives a great benefit to a young electrical engineer, who many times heard that Julius Mayer never used the word "energy"; that Archimedes and Galileo calculated all by mathematical method but had no signs for calculations, which can be used now; Newton, who opened infinitesimal quantities, nevertheless made all proofs by means of geometrical symbols as well as Euclid and all other scientists lived in period between them.

To understand Tesla it is necessary not only to read his works in original, but also to explain meaning of terms in his scientific understanding, especially the ciphered part of his archive, where the same notion is used in different contexts and has a lot of senses. Tesla's theory cannot be understood without a clear hypothesis and method of thinking, which is similar to his own.

Advanced modern science gradually begins to study Tesla's heritage. We do not speak about commercial or university aspects of scientific knowledge, but about summits of scientific thought unknown to wide intellectual community, about the knobs of separated net of worldwide global informational space for humankind, in which everybody is connected with the others by the channels of planetary telecommunicational net. **This connection is made by two main coincident parameters, conformity of ideas and intellectual level.** However, the thing that joins all is a subjective comprehension of Time as a basis for new civilization. All early civilizations originate only from their understanding of Space.

According to Tesla, in planetary community of the future all energy will be taken from inexhaustible and free sources. He pointed out that the Earth is a core (kernel) of a huge generator, which in rotation creates a difference of potentials equal to billions volts with a slowed ionosphere, that **humankind really lives in a spherical condenser of big capacity**, which continuously self-charges and self-discharges. In this condenser ionosphere is a "phase", atmosphere is a "dielectric" and the Earth is "zero". Thus, a global electrical process proceeds on our planet. Electrical energy does its work and then return to the natural environment of the planet Earth.

There are **linear "accelerators" by Tesla in the form of open vacuum tubes, i.e. tubes working at room temperature without energy losses**, since while laying at "target", electrostatic properties of the "target" itself are used; that's why it is possible to transmit any quantity of energy at any distance by means of induction. Kenneth Corum, who repeated Tesla's experiment in America and achieved some results, still **didn't understand the essence: Tesla's "particles" do not travel in space as other known particles or Hertz's waves, but they are self-generating from inductive field like "fire balls".**

The question is about an original Tesla's theory, which is related with cosmic radiation and was proved by him experimentally. Tesla's reversible magnetic field has a

universal significance. It is a mathematical idea, which is realized directly in every echelon of cosmic being.

I make bold to state that even Tesla's inductive motor, the so-called asynchronous motor, is not studied enough in conceptual sense. There is an undiscovered mystery of heavenly bodies rotation and rotation in general. Acceleration of motion of "Tesla's egg" is based on the change of field's geometry, but not on the properties of magnetic induction (actio in distans) or increasing of force (electricity input is permanent).

Among other Tesla's mysteries there are unstudied ones: in radio engineering there are some transmissions on the same frequency, which cannot be jammed (Twenty unused patents); in the structure of matter ("**I used to split the atoms, without releasing any energy**" – it is Tesla's statement of 1933); concerning attraction force there is a model of gravitational motor with tin engine and glass stator working once a year under the influence of certain planet arrangement; in the theory of ether (**matter is structured from ether and dissolves in ether again following simple mathematical laws. But if the generated energy is more than disappeared one, then cosmic catastrophes take place!**);

Tesla's medical devices and **influence of low-frequency waves on the brain operation**, which calls contractile motions and change of subjective "second".

This is Tesla's relation to Buddhist idea that the "I" is illusionary: "Really, we are something different, we are like a waves in subjective time and space, and when these waves disappear, nothing remains from us. There is no personality. We cannot saw that waves in the ocean have individuality. There is only an illusionary sequence of waves, which go one after another. We are not the same that was yesterday; I am only a sequence of relatively existences, which are not quite similar to each other. This sequence is that thing, which create an effect of continuity, like a movie, but it is not my subjective and mistaken understanding of my real life."

I cannot imagine Tesla smiling. I see him to be sad, since I see what a man could draw, when he gave all his essence to the power of natural laws.

MOST IMPORTANT DATES OF NIKOLA TESLA'S LIFE

1856 – on July 10, at the midnight Nikola Tesla was born in Smilyan, in Lika (province of Austrian monarchy) in the family of Serbian priest.

1875-1878 – he studied in Polytechnic Institute in Graze (Austro-Hungary).

1882 – in Budapest, have been walking around the park, Tesla made a discovery relating to the reversible magnetic field.

1883 – in Strasbourg he was employed in Edison's Continental Company and built a model of the first inductive motor.

1884 – he left for New York and began to work in Edison's laboratory.

1885 – he left Edison, established his own company and started to produce motors and generators of multiphase alternating current.

1888 – on May 16 he read a lecture in American institute of electrical engineers on the topic "New system of motors and transformers of alternating current".

1890 – he announced his results on physiological influence of high-frequency electricity.

1891 – he read lectures on the topic "Experiments with alternating current of very high frequency and its application in artificial wireless lighting"; registered patents of "spark oscillator with resonance transformer".

1892 – he came to London and read lectures in Royal scientific society on the topic "Light and other high-frequency phenomena", in the institute of electrical engineers he read lectures on the topic "Experiments with alternating current of high voltage and high frequency"; then the same lectures he read in Paris.

1893 – he attracted public attention on the World Exhibition in Chicago. He passed a high-frequency current through his body and demonstrated a model of reversible magnetic field, the so-called "Tesla's rotating iron egg".

1895 – on May 13 Tesla's laboratory in New York was burnt with fire.

1897 – he registered twelve patents in radio engineering.

1898 – registration of the patent for teleautomatic control (experiment with the ship on the lake in New York, Central Park. Tesla made a wireless control of the ship and staggered all witnesses).

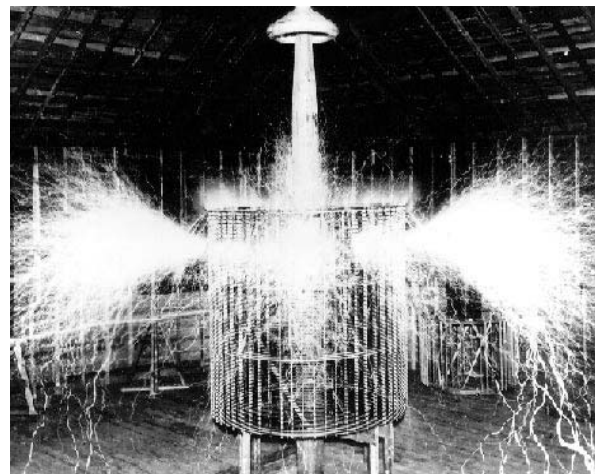
1899 – a construction of hydroelectric power station at Niagara was finished.

1899-1900 – experiments in Colorado Springs.

1901-1905 – construction of Wardenclyffe Tower at Long Island, near New York.

1909-1922 – he registered patents only in mechanical engineering (pumps, tachometers, bladeless turbines).

1943 – on January 7 Tesla died in New York, in "New Yorker" hotel. At first, he was read the burial service according to Orthodox tradition, and then his body was burnt according to Buddhist ceremony. *(My question is if he was truly burnt.... Or transfigured and conveyed...not as mere ashes... His main legacy is in Tesla Museum in Belgrade City, Yugoslavia. Prof. Velimir Abramovic.)*



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Future Energy Technologies

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Abstract

Today 85% of our country's energy comes from the combustion of dead fossils, a dirty fuel that is forcing the world's atmosphere to overheat. However, new 21st century energy sources that produce no carbon emissions and do not contribute to global warming are now emerging. Beyond the realm of fuel cells and hydrogen is the non-conventional world of "future energy." Some of the best examples are new and exciting generators that release trapped potential energy from nature in ways never dreamed of before. Others innovatively apply clean fuels in conventional systems that are surprisingly simple and yet very

efficient. Still others qualify as promising theoretical technologies that are a focus of attention for NASA and the USDOE. Most of them have one thing in common: they are very scientific but are relatively unknown to the general public. This presentation summarizes the latest breakthroughs in future energy. With scientific explanations of the input energy and output energy, the overunity efficiencies can be understood by average audience members. Included in the quantitative article are the inventions of Brown, Graneau, Jefimenko, Miley, Shoulders, Wallman, and others. The energy revolution is now beginning. It is time to understand the clean alternatives to dead, poisonous fuel.

Keywords: future energy, overunity, betavoltaic, biomass, atmospheric electricity, COFE

Introduction

In 1998, the U. S. Department of Energy (DOE) issued its *Comprehensive National Energy Strategy* (CNES)[1] that included as one of its five goals, the following aspiration:

Goal IV: Expand future energy choices – pursuing continued progress in science and technology to provide

future generations with a robust portfolio of clean and reasonably priced energy sources.

Objective 1. Maintain a strong national knowledge base as the foundation for informed energy decisions, new energy systems, and enabling technologies of the future.

Objective 2. Expand long-term energy options.

However, the DOE has not engaged in developing, much less maintaining a robust knowledge base of future energy choices, nor expanded research into new energy systems or long-term energy options, mainly due to upper management decisions. In a study performed by Integrity Research Institute on the progress of the CNES two years later, it is surprising that instead the DOE has worked to actively suppress enabling technologies of the future. Furthermore, concern for global warming and the expected increase in carbon emissions by the American society clearly do not enter DOE policies. In the past, the DOE has: (1) endorsed natural gas use for future generations, (2) rescinded a Nuclear Energy Research Initiative (NERI) grant awarded to a prominent professor for transmuting radioactive waste, and (3) reversed an initial offer to host a Conference on Future Energy (COFE). Therefore, it is clear by these and many other DOE practices that it is up to the private sector to conduct scientific research into new energy systems and enabling technologies of the future in order to replace carbon-emitting fuel systems. As a guideline, it is generally agreed that emerging energy technologies that qualify as true future energy must not produce carbon emissions nor contribute to global warming if we are to have a future planet earth. The reason for this is as Worldwatch Institute notes: "Stabilizing atmospheric CO₂ concentrations at safe levels will require a 60-80 percent cut in carbon emissions from current levels, according to the best estimates of scientists." [2] Another good reason is that we are running out of oil, according to the US Geological Survey, who offers a free educational poster online explaining the problem:

<http://geopubs.wr.usgs.gov/open-file/of00-320/of00-320.pdf>.

Future Energy Overunity

To understand emerging energy principles, it is helpful to examine the operation of a heat pump, which converts environmental free energy into useful work. The standard heat pump is a good example of an "overunity" system (energy out > energy in) releasing potential energy from the environment where the heat energy output is always in the range of 2 up to 7 times the input electrical energy. This so-called "coefficient of performance" represents an overunity efficiency, that does not violate any physics laws, if one considers, as the consumer does, how much energy must he put in to get the predicted energy output. Thus, the concept of "overunity," as also the concept of "free energy" has evolved from the consumer's point of view. What does it cost him to receive his heat, air conditioning, cleaning, or propulsion outputs? The closer it gets to "free," the

more desirable it is for the consumer and, we might add, to third world countries who cannot afford to build the thousands of miles of high voltage wires (infrastructure) to support a centralized energy system. Locally installed, modular heat and electricity generators will replace present utility-based service in the future. Then, large area blackouts will be a thing of the past. Energy will be for the most part, a one-time investment, included in the house, car, or spaceplane of one's choice. However, much needs to be done for these systems to supplant the established energy businesses that are the nation's major polluters. A commitment to a carbon-free energy economy, with financial backing, is required for such large changes to take place.

Cold Fog Discovery

As reported in the Journal of Plasma Physics,[3] the output energy thus exceeds the input energy by about 100% creating a 2-to-1 overunity condition

Many other systems exist today, in a research, development, or theoretical stage, which also convert potential energy into useful work. The first example is the "Cold Fog" invention of Dr. Peter Graneau from Northeastern University that converts chemical bond energy into kinetic energy. Inter-molecular bond energy in water is an available amount of energy estimated at 2.3 kJ/g. When injected with a high voltage capacitor discharge of 39.8 Joules, normal rainwater is accelerated into a cold fog that loses about 31.2 Joules of low-grade heat and a comparable amount (29.2 Joules) in fog kinetic energy output. As reported in the *Journal of Plasma Physics*, [3] the output energy thus exceeds the input energy by about 100% creating a 2-to-1 overunity condition favorable for reduction to a motorized conversion system.

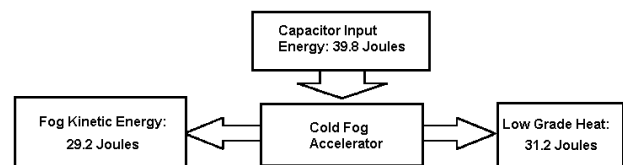


Fig. 1. Cold Fog Energy Flow

Betavoltaic Battery

The useful battery life is thus estimated to be about 25 years...

The next technology of importance is the betavoltaic battery invention of Dr. Paul Brown (U.S. Pat. #4,835,433). It involves a benign nuclear source called tritium (an isotope of hydrogen) that simply emits an

electron (5.7 keV beta particle) over its half-life of 12.5 years. The useful battery life is thus estimated to be about 25 years. It is a cheap, long-life, high energy density battery with a wide range of applications. Presently, Lucent Technologies has been contracted to produce the tritiated amorphous silicon for use in the semiconductor industry and even for watch batteries. The amorphous silicon is placed between two electrodes in order to complete the battery construction. The batteries have a mean energy density of 24 watts per kilogram and are ideal for low power, long-life applications [4]. It is clear that no recharging of these batteries is ever needed. ***The disposal is even safer than disposing of smoke detectors.***

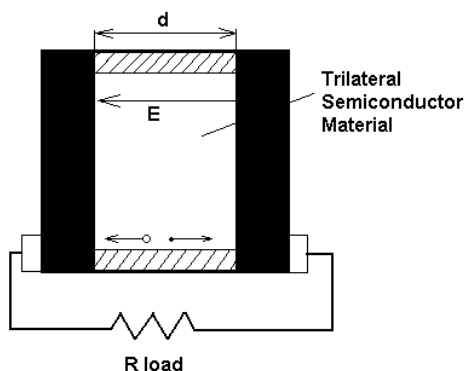


Fig. 2. Tritium Battery

Nuclear Remediation

It is worthwhile mentioning that Dr. Brown's other endeavor may give a boost to the nuclear power industry. He has discovered that low energy gamma rays (photons) on the order of 10 MeV, can function as an effective agent to transmute nuclear waste into short-lived isotopes, acceptable for burial anywhere. The remediation project is spearheaded by Nuclear Solutions, Inc. (NSOL) which plans to build a pilot plant to accept nuclear waste of any type and generate electricity at the same time. The Battelle Institute, Brookhaven Labs, and Los Alamos Labs have all been involved in the planning and testing stages of this new technology. Dr. Brown presented details of this invention at COFE [4]. Also, the State Department recently connected him with foreign markets that have assisted in proving its worth.

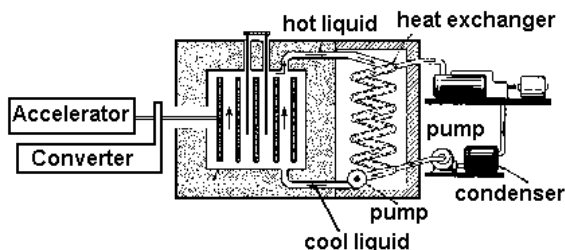


Fig. 3. Accelerator Driven Reactor

Electrostatic Motors

The next energy breakthrough is Dr. Oleg Jefimenko's electrostatic motors. Discovered by Ben Franklin in the

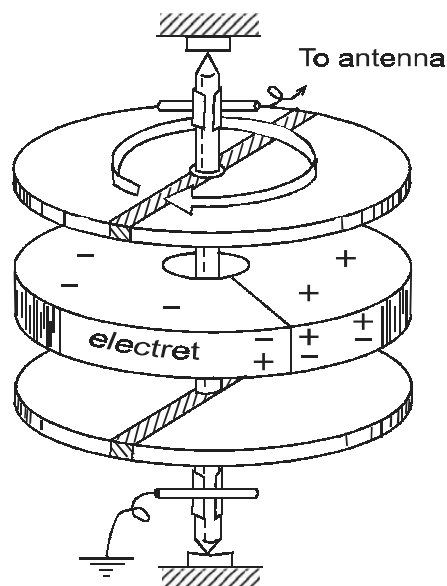


Figure 4. Electrostatic Motor Model

18th century, electrostatic motors are an all-American invention. They are based on the physics of the fair-weather atmosphere that has an abundance of positive electric charges up to an altitude of 20 km. However, the greatest concentration is near the ground and diminishes with altitude rapidly. Dr. Jefimenko discovered that when sharp-pointed antennas are designed for a sufficient length to obtain at least 6000 volts of threshold energy, the fair-weather current density available is about a picoampere per square meter. Such antennas produce about a microampere of current. However, small radioactive source antennas may be used instead that have no threshold voltage and therefore no height requirements. These antennas have larger current potentials depending upon the radioactive source used (alpha or beta source) and ionize the air in the vicinity of the antenna. Electrostatic motors are lighter than electromagnetic motors for the same output power since the motor occupies the entire volume. For example, it is expected that a motor one meter on a side will provide a power of one megawatt and weigh 500 kg or less.

Electrostatic motors also require very little metal in their construction and can use mostly plastic for example. They can also operate from a variety of sources and range of voltages. As Dr. Jefimenko points out, "It is clear that electrostatic motor research still constitutes an essentially unexplored area of physics and engineering, and that electrostatic motor research must be considered a potentially highly rewarding area among the many energy-related research endeavors." [5] The atmospheric potential of the planet is not less than 200,000 megawatts. He has succeeded in constructing demonstration motors that run continuously off atmospheric electricity. Jefimenko's largest output motor was an electret design that had a 0.1 Hp rating. [6] Certainly the potential for improvement and power upgrade exists with this free energy machine.

Biomass Gasification



Figure 5. Gasification Demo
Photo: Alternative Energy Institute

Clean fuels are difficult to find today. One example that satisfies a limited definition of "clean" is the carbo-hydrogen gas produced from biomass. David Wallman has patented the process for producing COH_2 from a high voltage discharge through any biomass solution (Pat. #5,417,817). This gas burns cleanly, producing water vapor and only the amount of CO_2 that was originally absorbed by the biological mass when it was growing in the ground. Contrast this with burning fossil fuels (oil and natural gas) which resurrect old buried carbon and add it to the atmosphere from ancient cemeteries in the ground. Instead, biomass gas burning recycles recently absorbed atmospheric carbon dioxide. The input energy is typically about a thousand watt-hours or about 3300 BTU to produce about 250 liters per hour of carbo-hydrogen (8.5 cubic feet per hour). With a heating value of over 500 BTU per cubic feet, the COH_2 output energy exceeds 4000 BTU, often approaching 5000 BTU in high efficiency designs. Thus, this biomass gasification process has an overunity efficiency of about 125% to 150%. However, when the entire energetics of the system are accounted for, including the ultraviolet light radiation, heat loss, etc., estimates of 200% to 400% are reasonable. Again, this process is a largely untapped resource while millions of gallons of farm-produced liquid biomass going to waste instead. Demonstrations of pilot plant designs are available from Wallman's company to replace present dependence on foreign oil (which is a fossil fuel). Municipal sewage treatment is a logical application for this invention. [7]

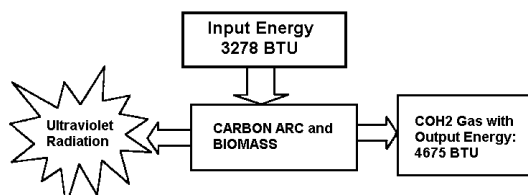


Figure 6. Biomass Gasification

Charge Clusters

An unusual energy source is the clustering of electrons by a discharge needle into a high density bundle equaling Avogadro's density of a solid[8]. Ken Shoulders has patented a process (Pat. #5,153,901) that produces electron clusters with such high energy density, they equal processes exceeding 25,000 degrees Celsius upon impact. Yet, he only uses 20 microjoules to produce the effects. The clusters travel at a maximum of one tenth of the speed of light and penetrate any substance with accuracy and sharp precision. It is similar to xenon clustering techniques currently used at megavolt energy levels. Low energy nuclear transmutation of the target has also been achieved with this process. Using a deuterium loaded palladium foil, only the bombardment areas show transmutation into silicon, calcium, and magnesium with electron clusters upon analysis with X-rays.

Fox has postulated that the high velocity electron clusters achieve results similar to ion accelerators, including penetration of the nucleus, with substantially less power. The new physics of like-charges clustering in bundles under low power conditions opens a wide range of applications including spacecraft maneuvering microthrusters. **The overunity efficiency is 9 to 1.**



Figure 7.

Charge cluster borehole into lead glass. Hole is about 10 micron diameter. Penetration is about 1mm per kV. The slowest speed clocked has been 1 cm in 50 nanoseconds. With an estimated 100 billion electrons carrying 100,000 positive ions, the kinetic energy exceeds 180 microjoules. It has been found that quantum mechanical forces overcome the Coulomb repulsion of like charges. (photo credit: Ken Shoulders)

Thin-Film Electrolytic Cell Power Unit

A product with the consumer in mind is Dr. George Miley's invention that produces about one watt per cubic centimeter of electrolyte [9]. Using a flowing packed-bed type electrolytic cell with 1-molar LiSO_4 in light water, small (1-mm diameter) plastic beads with a thin (500-1000 angstrom) film of metal (nickel, palladium, or titanium) are employed. A special sputtering technique to spray on the metal is used. With 2-3 volts of electrical power and only 1-5 milliamperes of current, the single film experiments produce an excess power ten times the input power! (The input

power is at most 0.01 watts while one half of a watt of heat is produced.) Observed power densities were 1 W/cc and above. It is also apparent that the physics of this reaction involve nuclear transmutations as well. As Dr. Miley notes: "The key finding from these studies has been the observation of a large array of "new" elements (i.e. different from the bead coating), many with significant deviations from natural isotopic compositions, after the run. Great care has been made to insure that these elements are distinguished from isotopic impurities by use of a "clean cell" with high purity components/electrolyte, in addition to the pre- and post-run analyses." Even low-energy radiation was detected from the beads days after

Future energy choices are already here...

each experiment. Application to space power, providing a 1-kW cell with only 500 cc of active electrode is predicted. Note that this particular invention, with its large overunity energy yield, was awarded a NERI grant by the DOE but then promptly withdrawn after certain individuals pressured the DOE into a re-evaluation of its grant to Professor Miley. The politics that override such grant decisions by the DOE Office of NEST are highly questionable.

Conclusion

Future energy choices are already here. In spite of the DOE lack of initiative in long range energy solutions, private inventors in this article have pioneered energy discoveries with a range of energy production possibilities. With Dr. Graneau's cold fog demonstrating a new energy source and a possible propulsion source, developmental efforts are ongoing with Hathaway Labs in Toronto to maximize the energy transfer to a useful machine for market. Dr. Brown's tritium battery is a milestone for long-term energy demand that is in production, while his nuclear remediation project is progressing rapidly. Dr. Jefimenko's electrostatic motors clearly demonstrate an available energy source yet untapped. Wallman's biomass gasification is ready to be developed on a large scale. Shoulder's charge clusters demonstrate extraordinary energy production on a microscopic scale with reasonable upscaling anticipated. Dr. Miley's electrolytic power unit also shows an extraordinary energy output, which deserves more research and development support.

Other inventors that meet the future energy criteria include Dr. Deborah Chung, from the State University of N.Y. at Buffalo, who has discovered "negative" resistance in carbon fibers[10]. Another, James Griggs, the inventor of the hydrosonic pump (Pat. #5,385,298), represents an overunity "apparatus for heating fluids" which even exhibits sonoluminescence (now marketed by HydroDynamics in Rome, Georgia). Dr. Paulo Correa also qualifies with his pulsed abnormal glow discharge (PAGD) energy conversion system[11]. It is our belief

that all of these inventions have the qualifications to be acceptable to energy futures. Also, theoretically and experimentally, there is growing support for a breakthrough in zero point energy conversion[12], which is the subject of more than one patent, the most recent being Dr. Frank Mead's patent #5,590,031. Furthermore, the extraction of energy and heat from the vacuum has also been proposed by Drs. Harold Puthoff and Daniel Cole[13]. Certainly, if only the 2.6% disruption in the oil flow from the Mid-East in 1999 can cause immediate chaos in the gasoline prices in this country, we desperately need to cut the umbilical cord strangling us. Therefore, a more robust energy development effort is required to help us make the transition from dangerous fossil fuels. A more stable, long-term energy future is possible with new energy sources like these discussed in this article.

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Instantaneous Electric Communication

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Nikola Tesla (1856-1943), an outstanding inventor, was and still remains one of the most mysterious persons in the history of electrophysics. Whereas the most scientists were moving together in direction of microparticles investigations, as the basis of matter structure and of nature itself, he was going in opposite direction. He had a keen interest in the investigation of electric charge of the Earth as a whole. He was looking for the ways to influence on it, to control its state and methods of its regulation.

Therefore, exactly, the most of his searches, experiments, the purpose of constructions and buildings, created according to his conceptions, cause perplexity and misunderstanding of scientists even in nowadays.

The most mysterious of his main experiments were made in USA after 1904. After Nikola Tesla death in 1943, all his diaries and records over a period from 1904 year had mysteriously disappeared. Probably they were stolen (it was known, what to take). Lost records could “cast light” on one of the most “strange” of his buildings in the form of the enough tall tower, on the top of which a specially created toroidal transformer was placed. This transformer could create there a huge electric potential up to the billion volts.

Nikola Tesla switched on this tower-device, what caused the fright and even panic in mind of people from nearby settlements. Of course! Because of very high electric potential there began air ionization, which spread very high to the atmosphere accompanying by the effect of color play. Such luminous, color-playing sky caused even a horror of people, who knew nothing about the experiment made and its goals. They did not guess that Tesla by means of the electric charge, created of the tower, was influencing on the electric charge of the Earth as a whole (about 600000 Coulomb). There was a global scale in Nikola Tesla's investigations.

There is no point in detailed analysis of the fact that the potential of the tower top influenced on the Earth charge. Interaction of charges-balls with the distortion of field lines, distortion-distribution of charge on their surfaces, induced charge, is beautifully described even in school physics textbooks. In Nikola Tesla investigations the

Earth had the role of one of the charged balls. It was possible by changing of charge on the tower to deform electric charge distribution on the whole Earth surface at once. This deformation (electric currents) could be fixed at once in every point of the Earth surface. It is alluring to use this effect for data transfer telecommunication, both on the Earth, and in space.

After such introduction the question “How does the system of instantaneous electric communication for any distance look like and work?” is still opened. First of all, the readers need to know, that such instantaneous communication is possible in principle. The proving it theoretical calculations, are rather difficult for popular interpretation. Some part of readers can take it on trust, and those who are most interested in can apply to works of Oleinik V.P. (quantum physics) the professor from Kiev Polytechnic University. At the minimum there are two necessary works: Oleinik V.P. “Faster-than-light transfer of a signal in electrodynamics. Instantaneous action-at-a-distance in modern physics” (Nova Science Publishers. Inc. New York. 1999) and Oleinik V.P. “Latest development of quantum electrodynamics: self-organizing electron, faster-than-light signals, dynamical heterogeneity of time.” (Physical vacuum and nature. 4. 3-17. 2000).

“PC” magazine has devoted a rather significant article entitled “Computers and teleportation” to V.P. Oleinik works, concerning instantaneous electric communication (“PC” #6, 2000). Note, that the author of the given article has also found the possibility of instantaneous electric communication, but by means of materialistic methods, absolutely different from Oleinik’s ones, what is most important – two different solutions point to the possibility of this communication. “PC” #6, 2000 in the article “Circles on fields” cited mathematical formulae of the structure of electron electro-magnetic field as an illustration (it refers to the Earth too) that the author of this article has got.

The most attentive readers of that article could notice, that one vector Hz absolutely “ignores” Special Theory of Relativity, since its mathematical expression does not include the velocity of light, whereas it presents in other vectors as a product of electric and magnetic conductivity. Magnetic line of this Hz vector goes to infinity and returns back from infinity. It surrounds the whole Universe. It is alluring to use exactly this (Hz) line for the **instantaneous** communication for any distance.

It is not so difficult to do it. In the Fig. 1 the easiest and most available for understanding line of the **instantaneous** electric communication is shown. A rotating charged dielectric ball (an “electron”, isn’t it?) is used as transmitter. The ball can be electrically charged up to the limit of charge flow-out into the ambient space. Around the charged rotating ball there appears electro-magnetic field, entirely analogous to the electro-magnetic field of the Earth (and of the electron too). The central magnetic line Hz goes to the infinity and returns

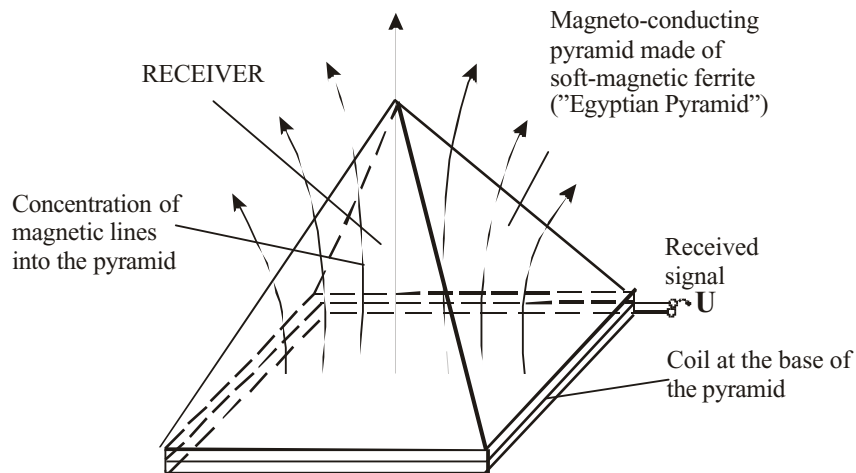
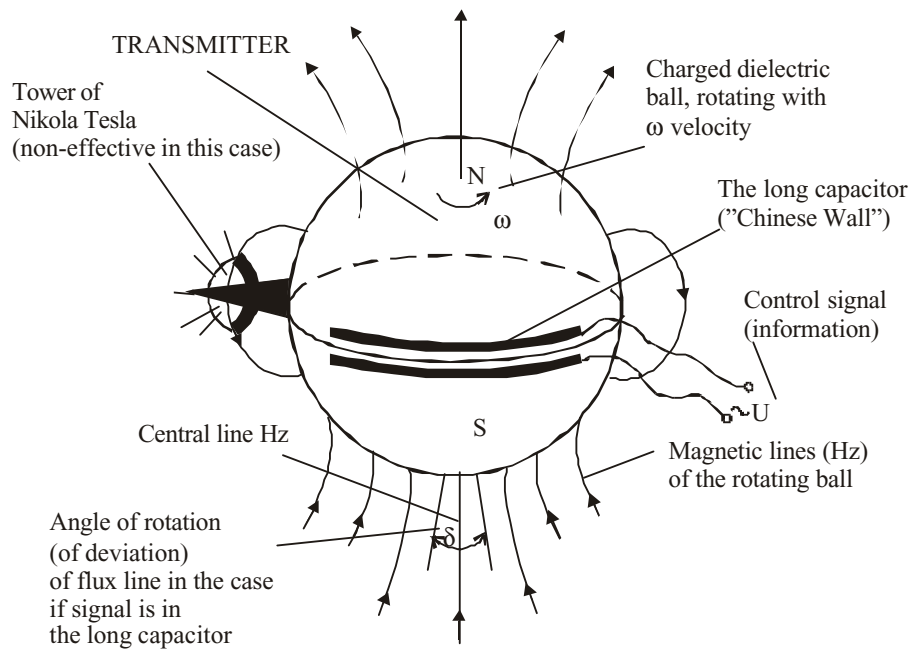


Fig.1

Line of the instantaneous electric communication on the basis of rotating charged ball and pyramid

back from it to the opposite side of the ball. In the same way the central magnetic line of the Earth (Hz) goes from one pole to the infinity through the whole Universe and returns from it to the center of the opposite pole.

If by the information to force the rotating ball (electron) to "wag by tail" (by Hz vector) which stretches through the whole Universe, then this "wagging" can be controlled instantly in every point of the Universe. While the rotating ball has a steady distribution of the surface charge, the line Hz does not change its dynamic position in the Universe.

If such distribution of the surface charge is broken, space position of the line Hz also will change. On mounting the Tower of Nikola Tesla on the surface of rotating ball and measuring the potential on this tower in time with an information it is possible to change the charge distribution on the ball, and, respectively, the space position of central magnetic line (Hz) in the whole Universe at once. Big disadvantage of the Tesla tower is that maximum influence on charge is executed in the point under the tower, and farther it began decreasing roughly (exponentially), according to physics laws.

Hence it is advisable to influence on the whole surface of charged ball, but not on some point of it. It is possible to influence at once on the very big part of surface by the long capacitor, placed on the perimeter (equator) of the charged rotating ball. Because of optimality reasons, this capacitor length should not exceed a quarter of the ball perimeter (equator) length. Charging and discharging this long capacitor on the ball equator by the data signal, only the position (angle ϕ) will be changed, not a value of the infinitely long magnetic line (Hz) in the Universe. It is a data transfer.

The natural question appears: "How to make on the Earth the most powerful transmitter for the **instantaneous** electric communication?" The answer suggests itself: "It is necessary to use the Earth itself as a rotating charged ball." It is not effective to use Nikola Tesla's Tower to deform the Earth electric charge. To place on the Earth very long (about thousands km) capacitor is quite easier. However, it must be placed not on the equator exactly, but moved a little bit because of the initial heterogeneity of the Earth surface charge distribution, caused by the presence of continents and oceans. It will be necessary to find the line of electric equator, where the amount of charge north and south of it is similar. This line will not be ideally straight and will be situated near the 30th parallel.

As a matter of fact, this grand capacitor is already built, but is half-broken. This capacitor is very well known – it is a Great Chinese Wall. The ancient, powerful Chinese Tzcin' Shi Huandi empire adapted and used it (capacitor) for the protection from nomads incursions. How unexpectedly and originally it is! In this case the electric iron would be the best tool for spiking. It is clear enough that the charged ball (as well as the Earth) will "wag by tail", which stretches through the whole Universe and does not change its energy, but only changes its position in space in time with information. Now we can go on to the question, concerning the way to control the Earth "wagging by tail" in the Universe, and thus to read information **instantaneously** in any point of the Universe.

In the Fig. 1 it is shown the input device of the electric communication receiver, made of the magnet sensitive material (it can be soft-magnetic ferrite) in the form of pyramid, with the proportions of well-known Egypt pyramids. Magnetic field lines of the far space pass through the pyramid from the top to the base and are concentrated by pyramid. If there is no signal (the "wagging by tail" of the far planet-transmitter is absent), then the magnetic flow, coming through the pyramid, does not change, and induced voltage in the coil, placed in the base of the pyramid, is absent (no information). If "wagging by tail" begins, then the magnetic flow, coming through the pyramid, will change, and it will cause the appearance of voltage on the coil in the base of the pyramid in time with the information.

Thus, the signal is received instantly. Here it is necessary to remind once again the difference between the

instantaneous and usual radio transmission. The usual radio transmitter for the transmission of the information uses the **energy** distortion of space by the information. This energy change in space happens with the velocity of light and hence there is the loss of time for information passing. In the considered case there is no **energy** change in space, there is only a change of magnetic lines position (Hz).

This is exactly the vivid and fundamental difference between the usual electric communication and the instantaneous one. **In other words, in usual transmitter during the fixed time interval there is the change of signal energy (instantaneous value), whereas in instantaneous transmitter there is no this change (only information). This is exactly the fundamental difference.**

Evidently, to receive instantly the signal from the opposite part of our Galaxy, we need rather big pyramid, in order to concentrate a big amount of field lines into the oscillatory circuit under the pyramid. The question can appear: why the pyramid, why not a cone? The point is that lines of the Earth magnetic field (the very lines that compass needle reacts on) in the any place of the pyramid horizontal section have the same density of distribution and are directed strictly parallel to the pyramid base. The cone in its horizontal section cannot provide such uniformity of distribution that is why it is not advisable to use it. From the space magnetic field lines pass through and concentrate in the pyramid strictly at right angle to the pyramid base.

This is the riddle of pyramids wonderwork. Any person coming into a pyramid, at the same moment feels the change of mental and physical condition of organism; which is very different from that it was before the entering into a pyramid. Of course! Visitors come inside, into concentrated magnetic field lines of the powerful and functioning magnetic core of the receiving electric circuit, what is absent outside the pyramid.

It is strange, but most of tourists are afraid of the ill effect, which can be produced by electrical systems on their health, but there they stand in a queue to feel this effect in pyramids. Concentration and division of magnetic field lines are the easy and effective way to reject a noise, created by the Earth magnetic field.

It is clear, pyramids should be oriented very thoroughly, so that lines of the Earth magnetic field would be strictly parallel to the base and to the opposite (East-West) sides of pyramid. To get such exactness of orientation in modern conditions is very problematically.

The most convenient place to build a pyramid (pyramids) is on the electric equator, in the place of its intersection with the electric meridian. Such place is located in Egypt, near its capital Cairo. And again we meet a paradox: such pyramids are already built on the Earth, but they are half-

broken. And Egypt was not less powerful than the ancient Chinese empire.

The Egyptian dynasty of Pharaohs has “completed” and adapted pyramids to burial-vaults, where mummies of dead Pharaohs were buried. Perhaps, it is even more incredible than in China. The impression is given that ancient powerful civilizations on the Earth had a competition between themselves, who will use radio-engineering constructions for instantaneous galactic communication in the most incredible way. Let’s give to a reader an opportunity to select a “winner”.

It must be noted that «PC» already published information that the Chinese Wall and Egyptian Pyramids are radio engineering constructions, intended for the instantaneous galactic communication (PC #114, 1997, etc).

There appears an interest in the possibility to produce very simple and manufacturable systems of instantaneous electric communication right now. Radio-electronic industry can produce them, but still does not guess about it.

Furthermore, such systems of instantaneous electric communication can be created at home, and even senior

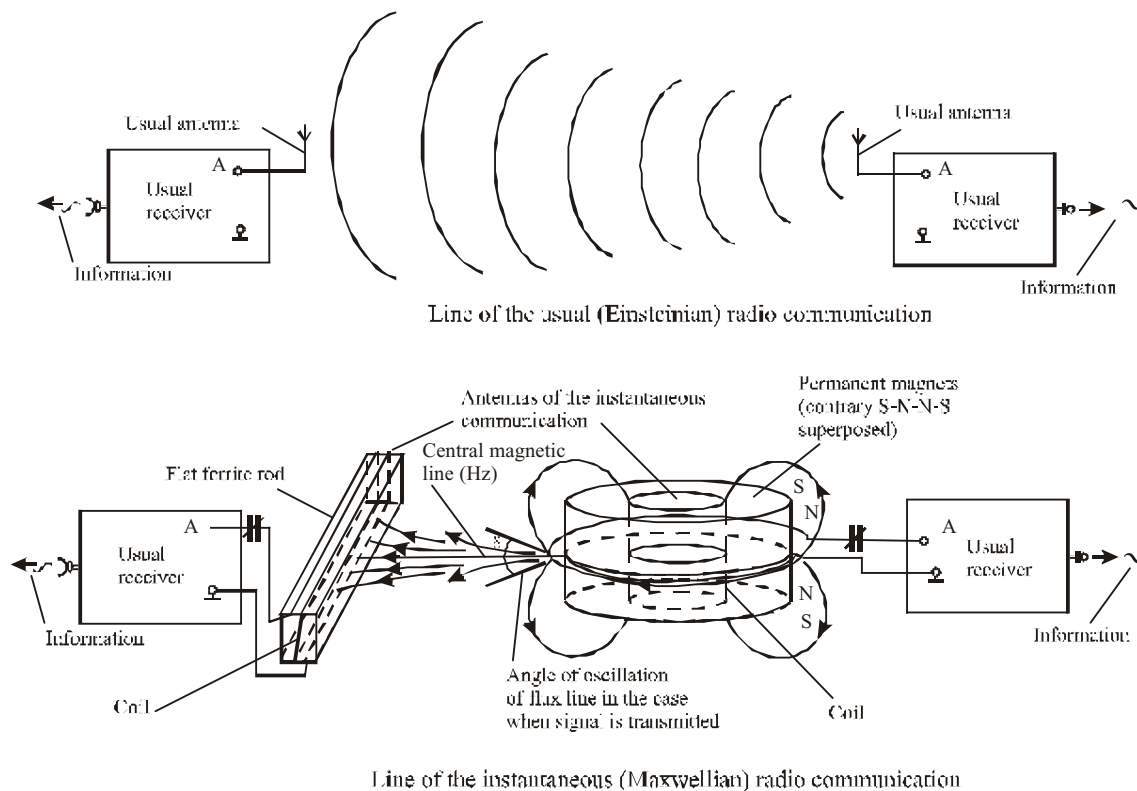


Fig. 2

Transformation of the usual line of communication of usual receiver-transmitter into the instantaneous one

pupils are capable to use them. In the Fig. 2 there is shown the construction of instantaneous electric communication line in comparison with the usual one. It can be produced even at home conditions. Two permanent magnets, connected between each other by analogous poles, are used as transmitting circuits.

Permanent magnets can be replaced by electromagnets. In the magnet connection point there is a coil, which while the signal passing through it will change its position (angle δ) in the space of the central magnetic line (Hz), coming out from the place of two magnets connection. Receiving circuit is available to be made of the flat ferrite, but coil

must be wound along, through butt-ends of core, so that the whole internal part of the coil would be maximal (in area extent) filled by ferrite.

The obtained coil can be completely “winded” (screened) by flat ferrite of big size. For more clearness of the experiment the central magnet line of transmitting part must be directed strictly along the axis of the receiving coil.

Now, if we give the alternating voltage (information) from transmitter to the transmitting coil, fixed on the permanent magnet, then the receiver, connected to the circuit of the

transmitting coil, placed on the flat ferrite, will detect an alternating voltage (information). Maximum effect is achieved at the resonance (coincidence of transmitter and receiver frequencies).

It is checked. It works. The dullest experts in radio-electronic (after the reading above) can rejoin, without making an experiment, that it is an absolute nonsense that any communication is out of the question. Coils with absolutely perpendicular axes, besides one of them is screened, do not interact with each other.

And here the most interesting thing starts. In the Fig.2 as it was mentioned above, the usual communication line and the instantaneous one were compared. The usual transmitter cannot generate the vector Hz that is why systems of usual and instantaneous communication cannot see each other in strict sense. What does it mean? It means, that in the same city it is possible to transmit on the one-carrier frequency ("what a nightmare!") two absolutely different television channels without any noises of one to another.

Usually by frequency match of a transmitter to the working frequency of another one, the radio communication is broken, but here it does not happen. Here some additional explanation should be given. As the vector Hz, which "ignores" the theory of Einstein is received from Maxwell equations, it follows that the usual (Einsteinian) system and the instantaneous (Maxwellian) one work on mutually perpendicular electromagnetic field lines (vectors).

In the Fig. 2 such difference is shown clearly. These are just "jokes" of complex numbers, when one value is absolutely perpendicular to another and nevertheless together they form a single whole. In other words it means, that two greatest persons in science Einstein and Maxwell as a matter of fact are something like "Siamese twins", completely grown together at the angle of 90 degrees, even by heads. On the one hand every one is on his own, but nevertheless they are the common (complex) organisms.

Hence there are a lot of misunderstandings on happened phenomena. How many scientists tried to find some mistakes of Einstein? They produced very convincing proofs concerning instantaneous interactions in nature. These scientists did not suspect that time and still do not guess now that they already for a long time are "walking" in the complex physics, which still does not exist. Einstein and Maxwell ("Siamese twins"), each occupies his own part of the complex number (complex physics) and they cannot be already taken off from there.

The only third, free "vacancy" is left to throw on the both of them at once the common "collar" and "reins", i.e. to fasten them ("twins") together by module and argument as any complex number. In this case no matter how the one part of complex number "ignore" the other one, only its argument will change, and module always will be equal

to the conditional unit. Only this single fact in principle changes the conception about "appearance" and "disappearance" of elementary particles! Even considered instantaneous and usual communications on the same receivers and transmitters in a complex conception (complex physics) eliminate appeared "misunderstandings" of all kinds.

Advantages of the instantaneous (Maxwellian) electric communication are especially evident during the connection with long-distance spacecraft. At present in the interval between sending of control signal to the station in region of Solar System peripheral planets and getting the reply it is possible to have a small break for dinner (it is very convenient).

In the case of the instantaneous electric communication use, duty operators will have "no dinner". Moreover, the system of instantaneous electric communication can realize two-way communication underwater and from underwater to overland. It is clear that input and output circuits of such system must be covered by slushing composite for the protection from aggressive effect of the salt sea-water. Such systems of instantaneous communication are very required to submarines.

Now, when readers know and understand the principle of operation of instantaneous (Maxwellian) electric communication systems and their advantages over usual ones (Einsteinian), we can only wait, when radio-electronic industry will start to produce these very required systems.

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NIKOLA TESLA'S LATER ENERGY GENERATION DESIGNS

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In the 1880's, Nikola Tesla invented the alternating current system we use today. By the 1890's, he was working on a new type of electrical generator that would not "consume any fuel".

This paper documents where in his writings the description of this new generator is found, a theory of how a fuelless generator could work and a suggestion as to how Tesla's new device might have operated.

ABSTRACT

Ten years after patenting a successful method for producing alternating current, Nikola Tesla claimed the invention of an electrical generator that would not "consume any fuel". Such a generator would be its own prime mover. Two of Tesla's devices representing different stages in the development of such a generator are identified.

INTRODUCTION

While in college Nikola Tesla claimed it should be possible to operate an electrical motor without sparking brushes. He was told by the professor that such a motor would require perpetual motion and was therefore impossible. In the 1880's he patented the alternating current generator, motor, and transformer.

During the 1890's he intensively investigated other methods of power generation including a charged particle collector patented in 1901. When the *New York Times* in June of 1902 carried a story about an inventor who claimed an electrical generator not requiring a prime mover in the form of an external fuel supply, Tesla wrote a friend that he had already invented such a device.

Fuelless electrical generation raises the same objection of perpetual motion, as did the generator in use today when it was first proposed. Research Nikola Tesla carried out during his second creative period and the resulting devices that were the basis for his assertion of fuelless electrical generation will be examined. Whether Tesla's fuelless generator was a "perpetual motion scheme" of the sort his teacher warned him against, or a creative application of recognized natural phenomena will be discussed.

TESLA'S STATEMENTS

In *The Brooklyn Eagle*, Tesla announced, on July 10th, 1931, "I have harnessed the cosmic rays and caused them to operate a motive device". Later on in the same article he said that "More than 25 years ago I began my efforts to harness the cosmic rays and I can now state that I have succeeded." In 1933, he made the same assertion in an article for the *New York American*, November 1st, under the lead in "Device to Harness Cosmic Energy Claimed by Tesla". Here he said:

This new power for the driving of the world's machinery will be derived from the energy, which operates the universe, the cosmic energy, whose central source for the earth is the sun and which is everywhere present in unlimited quantities.

Dating back "more than 25 years ago" from 1933 would mean that the device Tesla was speaking about must have been built before 1908. More precise information is available through his correspondence in the Columbia University Library's collection. Writing on June 10th, 1902 to his friend Robert U. Johnson, editor of *Century Magazine*, Tesla included a clipping from the previous day's *New York Herald* about a Clemente Figueras, a "woods and forest engineer" in Las Palmas, capital of the Canary Islands, who had invented a device for generating electricity without burning fuel. What became of Figueras and his fuelless generator is not known, but this announcement in the paper prompted Tesla, in his letter to Johnson, to claim he had already developed such a device and had revealed the underlying physical laws.

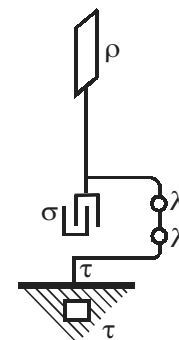


Fig.1

Particle collector

IDENTIFYING THE INVENTION

The device that, at first, seems to best fit this description is found in Tesla's patent for an "Apparatus for the Utilization of Radiant Energy", number 685,957, that was filed for on March 21, 1901 and granted on November 5, 1901. The concept behind the older technical language is a simple one. An insulated metal plate is put as high as possible into the air. Another metal plate is put into the ground. A wire is run from the metal plate to one side of a capacitor and a second wire goes from the ground plate to the other side of the capacitor. Then:

The sun, as well as other sources of radiant energy, throw off minute particles of matter positively electrified, which, impinging upon [the upper] plate, communicate continuously an electrical charge to the same. The opposite terminal of the condenser being connected to ground, which may be considered as a vast reservoir of negative electricity, a feeble current flows continuously into the condenser and inasmuch as the particles are ... charged to a very high potential, this charging of the condenser may continue, as I have actually observed, almost indefinitely, even to the point of rupturing the dielectric

(<http://www.xmission.com/~acparker/onichelson/>).

This seems like a very straightforward design and would seem to fulfill his claim for having developed a fuelless generator powered by cosmic rays, but in 1900 Tesla wrote what he considered his most important article in which he describes a self-activating machine that would draw power from the ambient medium, a fuelless generator, that is different from his Radiant Energy Device. Entitled "The Problem of Increasing Human Energy - Through the Use of the Sun", it was published by his friend Robert Johnson in *The Century Illustrated Monthly Magazine* for June 1900 soon after Tesla returned from Colorado Springs where he had carried out an intensive series of experiments from June 1899 until January of 1900.

The exact title of the chapter where he discusses this device is worth giving in its entirety:

A DEPARTURE FROM KNOWN METHODS -
POSSIBILITY OF A "SELF ACTING" ENGINE OR
MACHINE, INANIMATE, YET CAPABLE, LIKE AN
LIVING BEING, OF DERIVING ENERGY FROM THE
MEDIUM - THE IDEAL WAY OF OBTAINING
MOTIVE POWER

Tesla stated he first started thinking about the idea when he read a statement by Lord Kelvin who said it was impossible to build a mechanism capable of abstracting heat from the surrounding medium and to operate by that heat. As a thought experiment Tesla envisioned a very long bundle of metal rods, extending from the earth to outer space. The earth is warmer than outer space so heat would be conducted up the bars along with an electric current. Then, all that would be

needed is a very long power cord to connect the two ends of the metal bars to a motor. The motor would continue running until the earth was cooled to the temperature of outer space. "This would be an inanimate engine which, to all evidence, would be cooling a portion of the medium below the temperature of the surrounding, and operating by the heat abstracted" (<http://www.xmission.com/~acparker/onichelson/>), that is, it would produce energy directly from the environment without "the consumption of any material".

(No Model)

N. TESLA.

DYNAMO ELECTRIC MACHINE.

No. 406,968.

Patented July 16, 1889.

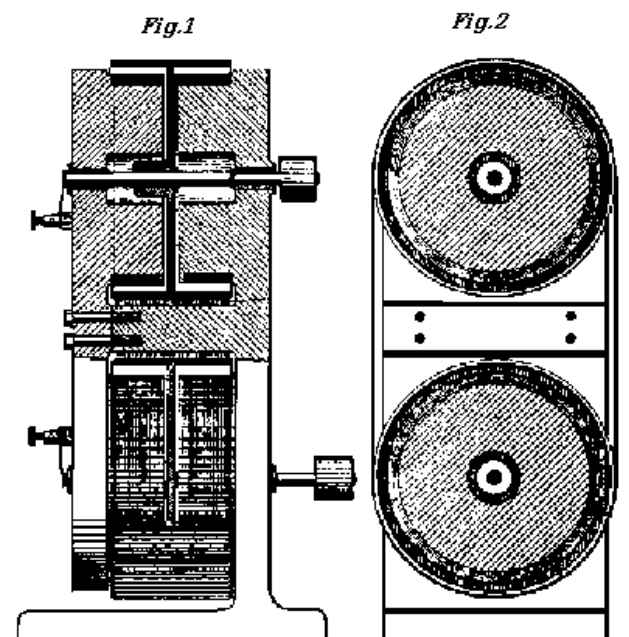


Fig.2

Dynamo Electric Machine

Tesla goes on in the article to describe how he worked on the development of such an energy device, and here it takes a bit of detective work to focus on which of his inventions he meant. He wrote that he first started thinking about deriving energy directly from the environment when he was in Paris during 1883, but that he was unable to do much with the idea for several years due to the commercial introduction of his alternating current generators and motors. It was not "until 1889 when I again took up the idea of the self-acting machine"

(<http://www.xmission.com/~acparker/onichelson/>).

THE TURBINE

He quickly came to realize that an ordinary electrical machine, like his generator, would not be able to directly extract energy from the cosmos and turned his efforts to what he called a "turbine" design.

The best-known turbine, that is, water pump, associated with Tesla is his patent for such a device,

#1,061,206, which was filed for in 1909 and granted in 1913. The unique point about this water pump is that instead of using some form of paddle wheels inside a box to move the water, he discovered that more water could be moved faster by using a set of flat metal disks. The turbine is, in itself, fascinating and may yet prove to be another important overlooked invention, but what is of concern regarding the electrical design is the general shape of the turbine - metal disks turning inside a supporting box.

This same shape turns up in another patent, this one for a "Dynamo-Electric Machine". This patent was filed and granted in the same year that Tesla said he returned to work on the "self-activating" machine, in 1889. The dynamo consists of metal disks that are rotated between magnets to produce an electric current.

Compared to his alternating current generator, this "dynamo" represents something of a curious throwback to the days of Faraday's early experiments with a copper disk and a magnet. Tesla makes some improvement over the Faraday setup by using magnets that completely cover the spinning metal disks and he also adds a flange to the outside of the disks so current can be taken off more easily - all of which makes for a better generator than Faraday's. On the surface, though, it is hard to see why Tesla patented such an anachronistic machine at this point in his work.

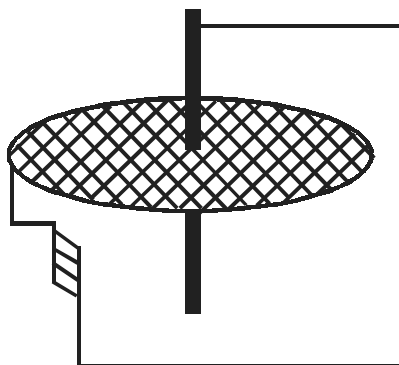


Fig.3

Faraday Generator with electromagnet

The next piece of the puzzle is found in an article Tesla wrote for *The Electrical Engineer* in 1891 entitled "Notes on a Unipolar Dynamo". Here Tesla presents an in-depth analysis of the Faraday disk generator, explains why it was an inefficient generator, describes his improved variations on the Faraday machine, and, at the bottom of the third page of the article, states that he has devised a generator in which "the current, once started, may then be sufficient to maintain itself and even increase in strength" (<http://www.xmission.com/~acparker/onichelson/>). Then, at the close of the article, he states that "several machines ... were constructed by the writer two years ago..." (<http://www.xmission.com/~acparker/onichelson/>). Two years before the writing of that article was 1889. All the evidence points to the turbine-shaped Unipolar

Dynamo as being Tesla's first design for a machine that can continue to produce electricity after being disconnected from an outside source of power.

SELF-SUSTAINING CURRENT

Before going into the details of this invention it would be worthwhile to have an idea of how any generator, even in theory, could be capable of producing a self-sustaining current. This has been clearly explained by Walter M. Elsasser in a *Scientific American* article (May 1958) titled "The Earth as a Dynamo".

Elsasser models the earth-dynamo, conveniently for this explanation, on the Faraday generator of a metal disk spinning over a bar magnet placed at the edge of the disk. He notes, also, that the bar magnet could be replaced by an electromagnet which could get its power from the spinning disk by attaching one end of the electromagnet's wire to the outside of the disk and the other end of the wire to the metal rod running through the center of the disk.

Elsasser then points out that an ordinary disk generator "could not maintain a current for very long because the current induced in the disk is so weak that it would soon be dissipated by the resistance of the conductor (the disk)". This conventional arrangement would not be an answer to "how currents could be built up and perpetuated to maintain the earth's magnetic field". He does, though, propose three options in the dynamo model that would explain the earth's persistent magnetism.

If we had a material that could conduct electricity a thousand times better than copper, the system would indeed yield a self-sustaining current. We could also make it work by spinning the disk very fast... a third way we could make such a dynamo self-sustaining ... is to increase the size of the system: theory says that the bigger we make such a dynamo, the better it will function. If we could build a coil-and-disk apparatus of this kind of scale of many miles, we would have no difficulty in making the currents self-sustaining

(<http://www.xmission.com/~acparker/onichelson/>).

Tesla did not have a material a thousand times more conductive than copper, neither was he able to spin a disk at the ultra-high speeds needed to produce such a current, nor did he plan on using a piece of rotating metal several miles in diameter. What he did was to use energy that is usually wasted in a generator and turn it into a source of power.

UNIPOLAR DYNAMO

Tesla's design varied from that of Faraday in two major ways. First, he used a magnet that was bigger in diameter than the disk so that the magnet completely covered the disk. Second, he divided the disk into sections with spiral curves radiating out from the center to the outside edge.

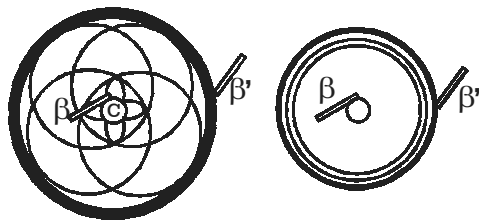


Fig.4
Tesla's Unipolar Generator

In the Faraday unipolar generator "the current", as Tesla noted, "set up will therefore not wholly pass through the external circuit ... and ... by far the greater portion of the current generated will not appear externally..." (<http://www.xmission.com/~acparker/onichelson/>). By having the magnet completely cover the disk, Tesla made use of the whole disk surface in current generation instead of only a small section directly adjacent to the bar magnet, as happened in the Faraday device. This not only increases the amount of current generated, but, by making the current travel from the center to the outside edge, makes all of that current accessible to the external circuit.

More importantly, these modifications on the Faraday design eliminated one of the biggest problems in any physical system - the reaction to every action. It is this reaction that works to cancel out whatever effort goes into causing the original action. In an electrical system if there are two turns of wire wound next to each other and a current is sent through the wire, the current passing through the first loop will set up a magnetic field that will work against the current passing through the second loop.

The spiral divisions in the disk cause the current to travel the full radius of the disk or, as in his alternative version of the generator, to make a full trip around the outside edge of the disk. Because the current is flowing in a large circle at the rim of the disk, the magnetic field created by the current not only does not work against the field magnet above the circular plate, as in conventional generators, but it actually reinforces the magnet. So as the disk cuts the magnetic lines to produce a current, the current coming off of the disk strengthens the magnet, allowing it to produce even more current.

Like conventional direct current generators, the unipolar dynamo also functions as a motor if current is put into the disk while under the magnet, and this seems to be the last element that could make the device self-sustaining, that is, capable of generating a current after being disconnected from an outside source of movement like falling water or steam.

Rotation is started by, say, a motor powered by line current. Both a generator and a motor disk are mounted in the magnetic enclosure. As the disks gain speed, current is produced which, in turn, reinforces the magnets, which cause more current to be generated.

That current is, likely, first directed to the motor disk, which increases the speed of the system. At a certain point the speed of the two disks is great enough that the magnetic field created by the current has the strength to keep the dynamo/motor going by itself.

What process might have kept the unipolar dynamo operating after the powered start-up is speculation at this point, however two features of the generator are significant. First, when a resistive load, like a light bulb is added to the circuit, it lowers the voltage at the center of the disk. This lower voltage at the center means that there is a greater difference in voltage between the center and the outside edge of the disk than there was before the light bulb was added. As the difference between the center and the outside increases, the dynamo works harder and makes more current. Second, yet more important, the dynamo takes either very little, or no energy to keep going because the current coming off the generator is doing double duty. The current makes the bulb glow, but on its way from the generator to the filament in the bulb, it travels a path that adds to the momentum of the dynamo and, therefore, consumes energy at a very low rate. The process continues, it would seem, until heat losses in the filament equal the rotational energy of the generator's flywheel.

In terms of Elsasser's criteria for a self-sustaining generator, the Tesla unipolar dynamo comes closest to satisfying the condition of a better electrical conductor. It is not that a new material is used, but a new geometry is applied so that the current does not create its own opposing forces. This is similar, but not equivalent, to having a better conductor.

Whether or not the dynamo is in fact a "fuelless" generator it appears to be an ingenious feat of engineering that takes one of the basic principles of nature, an equal and opposite action for every action, and turns it, by the use of a novel circuit geometry, into a reaction that is additive to the original action. Instead of the opposite reaction slowing down the system that created it, the reaction adds energy to the system.

Tesla, however, was not satisfied with his mechanical self-sustaining generator. The dynamo would provide the energy to run a single machine, but his vision was to light cities and in the 1900 *Century* magazine article he elaborated on the theory of such a machine.

Imagine, he suggested, an enclosed cylinder with a small hole in it near the bottom. Let us say that this cylinder, he added, contains very little energy but that it is placed in an environment that has a lot of energy. In this case, energy would flow from the outside environment, the high energy source, through the small opening at the bottom of the cylinder, and into the cylinder where there is less energy. Also suppose that as the energy passing into the cylinder is converted into another form of energy as, for example, heat is converted into mechanical energy in a steam engine. If it were possible to artificially produce such a "sink" for

the energy of the ambient medium then "we should be enabled to get at any point of the globe a continuous supply of energy, day and night"

(<http://www.xmission.com/~acparker/onichelson/>).

He continued, in the article, to elaborate on his energy pump but changed the image slightly. On the surface of the earth we are at a high energy level and can imagine ourselves at the bottom of a lake with the water surrounding us equal to the energy in the surrounding medium. If a "sink" for the energy is to be created in the cylinder, it is necessary to replace the water that would flow into the tank with something much lighter than water. This could be done by pumping the water out of the cylinder, but when the water flowed back in, we would only be able to perform the same amount of work with the inflowing water as we did when it was first pumped out. "Consequently nothing would be gained in this double operation of first raising the water and then letting it fall down".

Energy, though, can be converted into different forms as it passes from a higher to a lower state. He said, "assume that the water, in its passage into the tank, is converted into something else, which may be taken out of it without using any, or by using very little power" (<http://www.xmission.com/~acparker/onichelson/>). For example, if the energy of the ambient medium is taken to be the water, oxygen and hydrogen making up the water are the other forms of energy into which it could change as it entered the cylinder.

Corresponding to this ideal case, all the water flowing into the tank would be decomposed into oxygen and hydrogen ... and the result would be that the water would continually flow in, and yet the tank would remain entirely empty, the gases formed escaping. We would thus produce, by expending initially a certain amount of work to create a sink for... the water to flow in, a condition enabling us to get any amount of energy without further effort

(<http://www.xmission.com/~acparker/onichelson/>).

Tesla recognized that no energy conversion system would be perfect, some water would always get into the tank, but "there will be less to pump out than flows in, or, in other words, less energy will be needed to maintain the initial condition than is developed (by the incoming water), and this is to say that some energy will be gained from the medium"

(<http://www.xmission.com/~acparker/onichelson/>).

He found that this pumping could be done with a piston "not connected to anything else, but was perfectly free to vibrate at an enormous rate"

(<http://www.xmission.com/~acparker/onichelson/>).

This he was able to do with his "mechanical oscillator", a steam-driven engine used for producing high frequency currents. The faster the pump would work, the more efficient it would be at extracting energy from the cosmos. Research along this line culminated in the

oscillator demonstrated at the Chicago World's Fair in 1893. It was not until much later, in the 1900 article, he revealed: "On that occasion I exposed the principles of the mechanical oscillator, but the original purpose of this machine is explained here for the first time"

(<http://www.xmission.com/~acparker/onichelson/>).

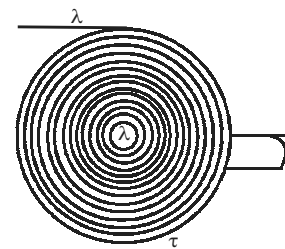


Fig.5
Coil for Electro Magnets

It was also in 1893 that Tesla applied for a patent on an electrical coil that is the most likely candidate for a non-mechanical successor of his energy extractor. This is his "Coil for Electro-magnets", patent #512,340 (Fig.5) It is another curious design because, unlike an ordinary coil made by turning wire on a tube form, this one uses two wires laid next to each other on a form but with the end of the first one connected to the beginning of the second one.

In the patent Tesla explains that this double coil will store many times the energy of a conventional coil

(<http://www.xmission.com/~acparker/onichelson/>). Preliminary measurements of two helices of the same size and with the same number of turns, one with a single, the other with a bifilar winding, show differences in voltage gain (<http://www.xmission.com/~acparker/onichelson/>). In Fig. 6, the upper curve is from the Tesla design; the lower was produced by the single wound coil. The patent, however, gives no hint of what might have been its more unusual capability.

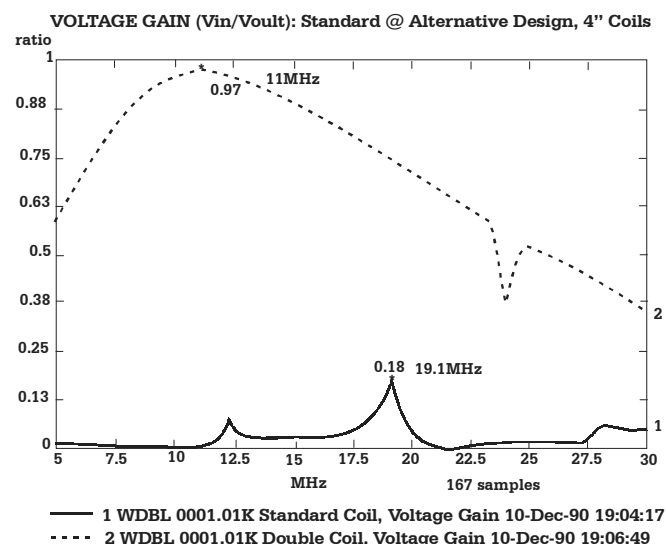


Fig.6
Voltage gain comparison

In the *Century* article Tesla compares extracting energy from the environment to the work of other scientists who were, at that time, learning to condense

atmospheric gases into liquids. In particular he cited the work of a Dr. Karl Linde who had discovered what Tesla described as a "self-cooling" method for liquefying air. As Tesla said, "This was the only experimental proof which I was still wanting that energy was obtainable from the medium in the manner contemplated by me" (<http://www.xmission.com/~acparker/onichelson/>).

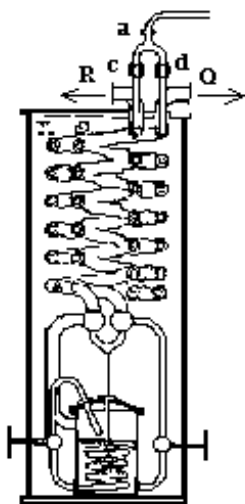


Fig.7
Linde's condenser

What ties the Linde work with Tesla's electromagnet coil is that both of them used a double path for the material they were working with. Linde had a compressor to pump the air to a high pressure, let the pressure fall as it traveled through a tube, and then used that cooled air to reduce the temperature of the incoming air by having it travel back up the first tube through a second tube enclosing the first

(<http://www.xmission.com/~acparker/onichelson/>). The already cooled air added to the cooling process of the machine and quickly condensed the gases to a liquid.

Tesla's intent was to condense the energy trapped between the earth and its upper atmosphere and to turn it into an electric current. He pictured the sun as an immense ball of electricity, positively charged with a potential of some 200 billion volts. The earth, on the other hand, is charged with negative electricity. The tremendous electrical force between these two bodies constituted, at least in part, what he called cosmic energy. It varied from night to day and from season to season but it is always present.

The positive particles are stopped at the ionosphere and between it and the negative charges in the ground, a distance of 60 miles; there is a large difference of voltage - something on the order of 360,000 volts. With the gases of the atmosphere acting as an insulator between these two opposite stores of electrical charges, the region between the ground and the edge of space traps a great deal of energy. Despite the large size of the planet, it is electrically like a capacitor, which keeps positive and negative charges apart by using a non-conducting material as an insulator.

The Earth has a charge of 90,000 coulombs. With a potential of 360,000 volts, the earth constitutes a capacitor of 0.25 farads (farads = coulombs/volts), (<http://www.xmission.com/~acparker/onichelson/>).

If the formula for calculating the energy stored in a capacitor ($E = 1/2 CV^2$) is applied to the earth, it turns out that the ambient medium contains $1.6 \cdot 10^{11}$ joules or 4.5 megawatt-hours of electrical energy.

In order to tap this energy storehouse Tesla had to accomplish two things - make a "cold sink" in the ambient energy and devise a way of making the "sink" self-pumping. Explaining how this process might have worked requires, again, speculation.

Such a "sink" would have to be at a lower energy state than the surrounding medium and, for the energy to continually flow into it, the "sink" would have to maintain the lower energy state while meeting the power requirements of the load attached to it. Electrical energy, watt-seconds, is a product of volts x amps x seconds. Because the period of oscillation does not change, either voltage or current has to be the variable in the coil's energy equation.

In that the double wound coil maximizes the voltage difference between its turns, it is probable that it is the current that is minimized to produce a low energy state in the coil. For the coil to be initially "empty" and at low energy would mean it operated at high voltage with a small amount of charge

(<http://www.xmission.com/~acparker/onichelson/>).

The coil, then, would be set into oscillation at its resonant frequency by an external power source. During a portion of its cycle the coil will appear to the earth's electric field as one plate of a capacitor. As the voltage across the coil increases, the amount of charge it can "sink" from the earth's higher energy field will increase. The energy taken into the coil - through the "small opening" which appears to be the atomic structure of the conductor according to the physics of Tesla's time - is "condensed" into positive and negative components of current, a lower energy state relative to the originating field.

The current is equivalent to the water converted to gases in Tesla's description of the self-acting engine. The current would "escape" from the "sink" into whatever load was connected across the coil. The movement of current into the load would produce a strong magnetic field (the stated intention of the patent) which, when it collapsed, would, again, produce a high potential, low charge "sink" to couple with the earth's electric field.

Because the inflowing energy performs a double function similar to the unipolar generator, supplying current to the load and aiding the pumping function, the system's energy expenditure in moving charge is low, allowing the system to gain more energy from the

medium than it expends in its operation. The coil needs no extra energy from an outside source to pump the energy it has extracted.

Energy would come directly from the sun

A more modern view of such a device, should it prove to operate in this theoretical manner, would be to describe it as a self-oscillating capacitive system. Once the device is set into oscillation, very little power is expended in driving the load. Because it is an electrostatic oscillating system, only a small amount of charge moves through the load per cycle, that is, the coulomb per seconds = amps are low. If the charge is used at a low rate, the energy stored in the capacitive system will be turned into heat at a slow rate enabling the oscillations to continue for a long period of time.

With his prominent position in the world of science at the time, it is curious why Tesla's invention was not commercialized or at least publicized more. Economics, not science, appear to have been the main factor. The adoption of alternating current was opposed by powerful financiers of the period. Michael Pupin, another leading electrical researcher at the turn of the century, noted in his autobiography:

...captains of industry...were afraid that they would have to scrap some of their direct current apparatus and the plants for manufacturing it, if the alternating current system received any support ... ignorance and false notions prevailed in the early nineties, because the captains of industry paid small attention to highly trained scientists
(<http://www.xmission.com/~acparker/onichelson/>).

Tesla's patents for electrical generators and motors were granted in the late 1880's. During the 1890's the large electric power industry, in the form of Westinghouse and General Electric, came into being. With tens of millions of dollars invested in plants and equipment, the industry was not about to abandon a very profitable ten-year-old technology for yet another new one.

Tesla saw that profits could be made from the self-acting generator, but somewhere along the line he had pointed out to him the negative impact the device would have. At the end of the section in *Century* where he described his new generator he wrote:

I worked for a long time fully convinced that the practical realization of the method of obtaining energy from the sun would be of incalculable industrial value, but the continued study of the subject revealed the fact that while it will be commercially profitable if my expectations are well founded, it will not be so to an extraordinary degree
(<http://www.xmission.com/~acparker/onichelson/>).

Years later, in 1933, he was more pointed in his remarks about the introduction of his fuelless generator. In the *Philadelphia Public Ledger* of November 2nd, is an interview with Tesla under the headline "Tesla

'Harnesses' Cosmic Energy". In it he was "Asked whether the sudden introduction of his principle would upset the present economic system, Dr. Tesla replied, 'It is badly upset already'. He added that now as never before was the time ripe for the development of new resources".

It has been nearly a century since Nikola Tesla claimed a radically new method for producing electricity. The need for the development of new resources is greater now than at the end of the last century. Perhaps these overlooked inventions will make his vision of "increasing human energy through the use of the sun's energy" become a reality.

Acknowledgment

Thanks to Mr. John Ratzlaff of Millbrae, California for generously sharing a variety of Tesla material that helped make this paper possible.

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Free Energy

From Tesla's Wireless Ionospheric Electricity

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Overview

The *Wardenclyffe Tower Centennial (1903-2003)* is an opportunity to celebrate a monument to Nikola Tesla's visionary genius. Recently, a resurgence of interest from prominent physicists has focused on the unusual method of pulsing a broadband Tesla coil at a repetition rate of 8 Hz to resonate with the Earth's Schumann cavity [1]. Nikola Tesla, the father of AC electricity, is responsible for recognizing that an atmospheric and a terrestrial storage battery already exists everywhere on earth, for the benefit of mankind. This is perhaps the "wheelwork of nature" that Tesla was referring to ("*... it is a mere question of time when men will succeed in attaching their machinery to the very wheelwork of nature.*" – *Tesla addressing the Amer. Inst. of Elec. Eng., 1891*). A century later, only a handful of visionary scientists recognize the untapped renewable reservoir of terawatts of electrical power (3000 gigawatts) that sits dormant above us, waiting to be utilized.

Background

In 2001, the Bush-mandated National Transmission Grid Study (NTGS 2001) was designed to identify the major transmission bottlenecks across the U.S. and identify technical and economic issues resulting from these transmission constraints. With deregulation of U.S. utilities and the lack of jurisdiction for the Federal Energy Regulatory Commission (FERC), the U.S. is fighting an electrical energy crisis which right now, costs consumers hundreds of millions of dollars annually due to interregional transmission

congestion. There is no longer any economic incentive nor any FERC eminent domain for states to provide rights-of-way, besides the lack of Federal compensation to utilities to build new transmission lines.



Fig. 1

The 187-foot Wardenclyffe Tower in 1903 which stood unfinished for the next 14 years

Historically, the creation of electrical utilities was beset with scandal, such as the six years of Congressional hearings starting

in 1928 in which “thousands of pages of testimony revealed a systematic, covert attempt to shape opinion in favor of private utilities, in which half truths and at times outright lies presented municipal systems in a consistently bad light [2].”

Today, US AID funds the U.S. Energy Association to train utility representatives from the former Russian states on how to reliably monitor electricity usage and collect money from customers in their respective countries, while those economically challenged people struggle for sufficient wages.

At a Washington DC conference which this author attended, called “Implementing a National Energy Strategy: Breaking Down the Barriers” also sponsored by the US Energy Association (12/01), only the depressing news about unresolved US electricity headaches were discussed. Editor of *Energy Daily*, Llewelyn King finally concluded, “*We are using 19th century technology for electrical transmission.*”

He then called for a paradigm shift toward new technology and cited the “monster infrastructure problems” within the U.S. as compared to the developing countries.

A year later (June, 2003) the US DOE held an emergency meeting with utility heads as a natural gas crisis looms from the lack of diversification of new electrical power generation facilities. “Innovation in new technology and renewable sources are needed in the long term to improve the environment and meet rising demand,” summarized an *Investors Business Daily* editor about the crisis [3].

In November, 2002, the American Council for the United Nations University called for *wireless energy transmission* to circumvent the need for transmission lines as part of their “Millennium Project.”

In cooperation with the National Science Foundation (NSF), NASA, and the Electrical Power Research Institute (EPRI), the beaming of microwave energy and the creation of a world energy organization was seen to actively address the 2020 challenges to global electricity supply, especially in areas of massive urban concentrations [4].

In 1940, “the United States prided itself on using half the world’s electricity [5].

”Since 1980, the U.S. has also doubled its dependence on foreign oil and *doubled* its electrical transmission grid inefficiency. From 31 Quads (quadrillion BTUs) generated, a full *2/3 is totally wasted* in “conversion losses” with only about 11 Quads (3.7 trillion kWh) delivered to the end-user [6].

Instead of trying to build 2 power plants per week (at 300 MW each) for the next 20 years (only to have a total of additional 6 trillion kWh available by 2020), as the Bush-Cheney Administration wants to do, we simply need to *eliminate the 7 trillion kWh of conversion losses* in our present electricity generation modality.

History of Tesla’s Wireless Energy

The fateful decision in 1905 by J. P. Morgan to abandon Tesla’s Wardenclyffe Tower project on Long Island (after investing \$150,000), was a result of learning that it would be designed mainly for *wireless transmission of electrical power*, rather than telegraphy.

No more money was forthcoming for the project that Morgan initiated, even when the equipment cost alone cost about \$200,000. Morgan believed that he would “have nothing to sell except antennas (and refused) to contribute to that charity [7].”

Tesla tried and tried for years until in 1917 the U.S. government blew up the abandoned Wardenclyffe tower because suspected German spies were seen “lurking” around it. With Edison as his willing ally, Morgan even publicly discredited Tesla’s name, so that all of the five school textbook publishers of the time removed any reference to him.

Any wonder why even today, 100 years later, hardly anyone knows who Tesla is?

The rest of this article will present a physics and electrical engineering argument for a subsequently forgotten engineering alternative for energy generation and transmission.

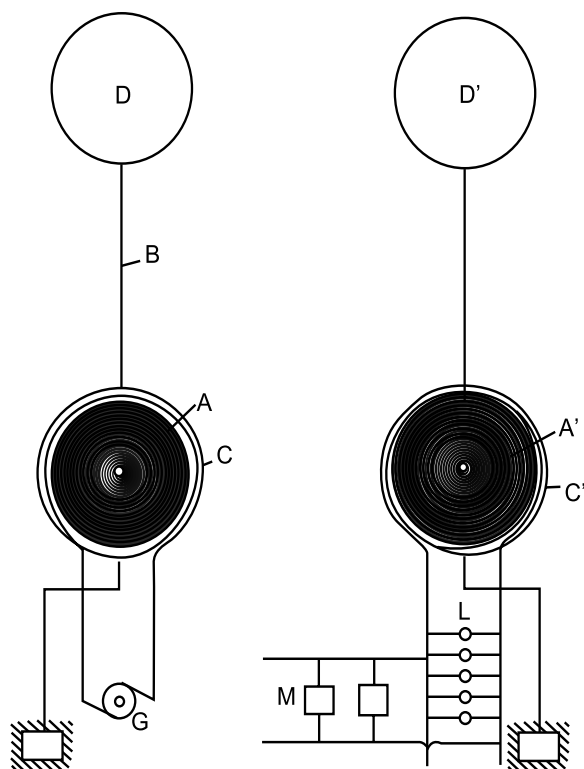


Fig. 2

As Tesla experimented with a 1.5 MW system in 1899 at Colorado Springs, he was amazed to find that pulses of electricity he sent out passed across the entire globe *returned with “undiminished strength.”* He said, “It was a result so unbelievable that the revelation at first almost stunned me [8].” This verified the tremendous efficiency of his peculiar method of pumping current into a spherical ball to charge it up before discharging it as a pulse of electrical energy, *a “longitudinal” acoustic-type of compression wave*, rather than an electromagnetic Hertzian-type of transverse wave. It was therefore, more akin to electrostatic discharge than wave mechanics.

Tesla also planned to include a *stationary resonant wave* creation globally, within the earth-ionosphere cavity, as part of the wireless transmission of power. Examining the pair of 1900 patents #645,576 and #649,621 each using the same figure on the first page, we find in the first patent that Tesla has designed a quarter-wave antenna (50 miles of secondary coil wire for a 200 mile long wavelength). More importantly is the sphere on the top which is supposed to be a conductive surface on a balloon raised high enough to be radiating in “rarefied air

(International patents, as well as US patents, are now available online from <http://gb.espacenet.com/espacenet/>.)”

As Tesla states, “That communication without wires to any point of the globe is practical with such apparatus would need no demonstration, but through a discovery which I made I obtained absolute certainty. Popularly explained it is exactly this: When we raise the voice and hear an echo in reply, we know that the sound of the voice must have reached a distant wall, or boundary, and must have been reflected from the same. Exactly as the sound, so an electrical wave is reflected, and the same evidence which is afforded by an echo is offered by an electrical phenomena known as a ‘stationary’ wave – that is, a wave with fixed nodal and ventral regions. Instead of sending sound vibrations toward a distant wall, I have sent electrical vibrations toward the remote boundaries of the earth, and instead of the wall, the earth has replied. In place of an echo, I have obtained a stationary electrical wave, a wave reflected from afar [9].”

Nikola Tesla’s discovery of pulsed propagation of energy does not resemble the standard transverse electromagnetic waves so familiar to electrical engineers everywhere. Many engineers and physicists have dismissed Tesla’s wireless energy transmission as unscientific without examining the unusual characteristics and benefits of longitudinal waves, which are the z-component solutions of Maxwell equations.

Tesla wrote, “That electrical energy can be economically transmitted without wires to any terrestrial distance, I have unmistakably established in numerous observations, experiments and measurements, qualitative and quantitative. These have demonstrated that it is practicable to distribute power from a central plant in unlimited amounts, with a loss not exceeding a small fraction of one per cent in the transmission, even to the greatest distance, twelve thousand miles – to the opposite end of the globe [10].”

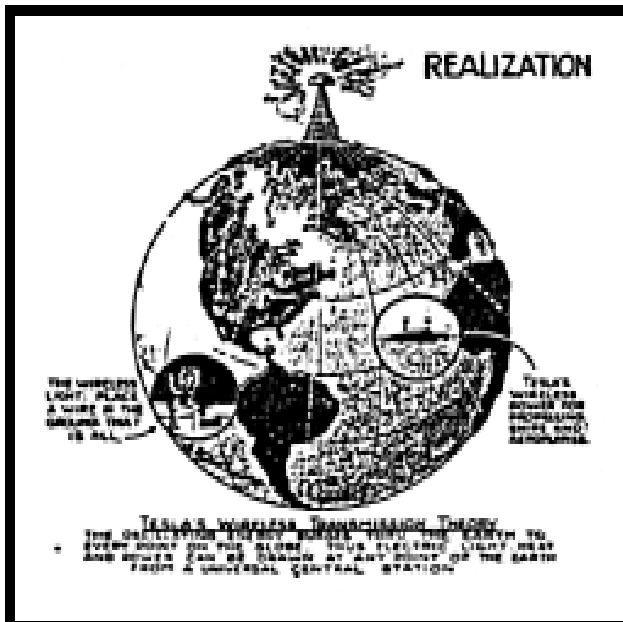
Tesla was an electrical genius who revolutionized our world with AC power in a way that DC power could never have accomplished, since the resistance of any transmission lines, (except perhaps, superconductive ones), is *prohibitive* for

direct current. He deserved much better treatment from the tycoons of his age, than to spend the last 40 years of his life in abject poverty. However, he was too much of a gentleman to hold a grudge. Instead, regarding the magnifying transmitter, Tesla wrote in his autobiography, “I am unwilling to accord to some small-minded and jealous individuals the satisfaction of having thwarted my efforts. These men are to me nothing more than microbes of a nasty disease. My project was retarded by laws of nature. *The world was not prepared for it. It was too far ahead of time. But the same laws will prevail in the end and make it a triumphal success* [11].”

Tesla's World System

Tesla's “World System” was conceptually based on three inventions of his:

1. *The Tesla Transformer* (Tesla coil)
2. *The Magnifying Transmitter* (transformer adapted to excite the earth)
3. *The Wireless System* (efficient transmission of electrical energy without wires).



Tesla states, “The first World System power plant can be put in operation in nine months. With this power plant it will be practicable to attain electrical activities up to 10 million horsepower (7.5 billion watts), and it is designed to serve for as many technical achievements as are possible without due expense [12].”

Tesla's calculated power levels are conservatively estimated and recently updated with contemporary physics calculations by Dr. Elizabeth Rauscher. For example, Professor Rauscher shows that the earth's ionosphere and magnetosphere contains sufficient potential energy, *at least 3 billion kilowatts (3 terawatts) respectively*, so that the resonant excitation of the earth-ionosphere cavity can reasonably be expected to increase the amplitude of natural “Schumann” frequencies, facilitating the capture of useful electrical power. Tesla knew that the earth could be treated as one big spherical conductor and the ionosphere as another bigger spherical conductor, so that together they have parallel plates and thus, comprise a “spherical capacitor [13].”

Rauscher calculates the capacitance to be about 15,000 microfarads for the complete earth-ionosphere cavity capacitor. In 1952, W.O. Schumann predicted the “self-oscillations” of the conducting sphere of the earth, surrounded by an air layer and ionosphere, without knowing that Tesla had found the earth's fundamental frequency fifty years earlier [14].

“All that is necessary,” says Dr. James Corum, is that Tesla's transmitter power and carrier frequency be capable of round-the-world propagation.” In fact, Tesla (in the *L.A. Times*, Dec. 1904) stated, “With my transmitter I actually sent electrical vibrations around the world and received them again, and I then went on to develop my machinery.” Dr. Corum notes in an article on the ELF (extremely low frequency) oscillator of Tesla's that the tuned circuit of Tesla's magnifying transmitter was the whole earth-ionosphere cavity [15].

Corum explains that a mechanical analog of the lumped circuit Tesla coil is an easier model for engineers to understand [16].

From a mechanical engineering viewpoint, the “magnifying factor” can be successfully applied to such a circuit. “The circuit is limited only by the circuit resistance. At resonance, the current through the circuit rises until the voltage across the resistance is equal to the source voltage. This circuit was a source of deep frustration to Edison because voltmeter readings taken around the loop did not obey Kirchoff's laws!” As a

result, Edison claimed such a circuit was only good for electrocution chairs.

Earth's Renewable Energy

Tesla's world system activates the earth's renewable electrical storage battery which normally sits dormant except during lightning strikes. Regarding simply the electrostatic energy storage capacity of the ionosphere, Dr. Oleg Jefimenko, author of *Electrostatic Motors*, explains that during one electric storm, the atmospheric electric field dissipates at least 0.2 terawatts (billion kilowatts), indicating that the entire earth must have even more total available energy [17].

Furthermore, the power loss experienced by Tesla's pulsed, electrostatic discharge mode of propagation was less than 5% over 25,000 miles. Dr. Van Voorhies states, "...path losses are 0.25 dB/Mm at 10 Hz," which often is difficult for engineers to believe, who are used to transverse waves, a resistive medium, and line-of-sight propagation modes that can dissipate 10 dB/km at 5 MHz [18].

The *capacitive dome* of the Wardenclyffe Tower, like the conductive balloon of Tesla's '576 patent, is a key to the understanding of the longitudinal waves. Dr. Rauscher quotes Tesla, "Later he compared it to a Van de Graaff generator. He also explained the purpose of Wardenclyffe...*'one does not need to be an expert to understand that a device of this kind is not a producer of electricity like a dynamo, but merely a receiver or collector with amplifying qualities* [19]."

Only a few great physicists like Drs. Elizabeth Rauscher, James Corum, and Konstantin Meyl have realized that Tesla was very practical when he proposed the resonant generation and wireless transmission of useful electrical power. (*Professor Konstantin Meyl sells a "Demo Set" that is a miniature dual dome like Tesla patent '576, a wireless longitudinal wave demonstration kit, available at http://www.k-meyl.de/Demo-Set/body_demo-set.html. Enter this link at www.freetranslation.com for English*). Tesla's knowledge of atmospheric

electricity transduction was so extensive and reliable that Jim Corum, who has been funded to continue Tesla's work, recently told me, "You just have to do exactly what Telsa did and you will consistently get the same results he did [20]."

After returning from his experiments at Colorado Springs in 1900, Nikola Tesla stated, "*If we use fuel to get our power, we are living on our capital and exhausting it rapidly. This method is barbarous and wantonly wasteful and will have to be stopped in the interest of coming generations* [21]."

In view of our present fossil-fuel-caused global warming, Tesla seems very prophetic from his vantage point of a century ago.

High Transmission Integrity and Low Loss

Tesla states, "As to the transmission of power through space, that is a project which I considered absolutely certain of success long since. Years ago I was in the position to transmit wireless power to any distance without limit other than that imposed by the physical dimensions of the globe. In my system it makes no difference what the distance is.

The efficiency of the transmission can be as high as 96 or 97 per cent, and there are practically no losses except such as are inevitable in the running of the machinery. *When there is no receiver there is no energy consumption anywhere. When the receiver is put on, it draws power. That is the exact opposite of the Hertz-wave system.* In that case, if you have a plant of 1,000 horsepower (750 kW), it is radiating all the time whether the energy is received or not; but in my system no power is lost. When there are no receivers, the plant consumes only a few horsepower necessary to maintain the vibration; it runs idle, as the Edison plant when the lamps and motors are shut off [22]."

These amazing facts are explained by Corum(s) and Spainol, "...the distinction between Tesla's system and 'Hertzian' waves is to be clearly understood. Tesla, and others of his day, used the term 'Hertzian waves' to describe what we call

today, energy transfer by wireless transverse electromagnetic (TEM) radiation...no one wants to stand in front of a high power radar antenna. For these, E and H are *in phase*, the power flow is a 'real' quantity (as opposed to reactive power), and the surface integral of $\mathbf{E} \times \mathbf{H}$ (Poynting vector) is nonzero.

In view of our present fossil-fuel-caused global warming, Tesla seems very prophetic from his vantage point of a century ago.
The case is not so simple in an unloaded power system, an RF transformer with a tuned secondary, or with a cavity resonator. In these situations, the fields are in phase quadrature, the circulating power is reactive and the average Poynting flux is zero – *unless a load is applied*. They deliver no power without a resistive load. These are clearly the power systems which Tesla created. The polyphase power distribution system was created by him in the 1880s and inaugurated at Niagara Falls in 1895. The RF transformer was invented and patented by him in the 1890s. Terrestrial resonances he experimentally discovered at the turn of the century. And, for the next 40 years he tried to bring through to commercial reality this global power system. Today, millions of us have working scale models of it in our kitchens, while the larger version sits idle [23].”

Note for a spherical, electrostatic pulse discharge, E is radial and H is helical since J is radial (longitudinal or irrotational current) [24].

Biological and Economic Impact

Another common criticism of the Tesla wireless power system is regarding its possible biological effects. Calculating the circulating reactive power, Corum(s) and Spainol find a density of a *microVAR per cubic meter* at 7.8 Hz, which is quite small, while it is well-known that such a frequency is very biologically compatible [25]. The authors also look at the present 100 V/m earth-ionosphere field and again find that raising it by a factor of 4 to 10 will

pose no ill effects. (Thunderstorms do it all of the time around the world.)

In terms of economic theory, many countries will benefit from this service. Only private, dispersed receiving stations will be needed. Just like television and radio, a single resonant energy receiver is required, which may eventually be built into appliances, so no power cord will be necessary! Just think: monthly electric utility bills from old-fashioned, fossil-fueled, lossy electrified wire-grid delivery services will be optional, much like “cable TV” is today. In the 21st century, “Direct TV” is the rage, which is an exact parallel of Tesla’s “Direct Electricity.”

Let us fulfill this prophesy of Tesla, making it a *triumphal success*, by supporting a philanthropic, international wireless power station installed on a remote island to electrify the whole world. The benefits, immediately making direct electricity available everywhere, are too numerous to count. (Coincidentally, in Tesla’s homeland, the Electric Power Company of Serbia raised their monthly rates *by 50 %* on the day *Harnessing the Wheelwork of Nature* was sent to the publisher, while California is now up to 15 cents per kWh, double the US average.)

Become educated about Tesla’s wireless energy transmission discovery at <http://users.erols.com/iri/tesla.htm> and help celebrate the Wardencllyffe Tower Centennial at the Tesla Energy Science Conference, November 8-9, 2003 (www.IntegrityResearchInstitute.org).

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About the Author

Thomas Valone received his Master's in Physics from the State University of NY at Buffalo (1984) and his Ph.D. in General Engineering from Kennedy-Western University (2003). He taught physics, AC electricity, microprocessors, digital logic and environmental science at Erie Community College in NYS (1982-1987), and is the author of several books and about 100 articles and reports. Presently, Dr. Valone is President of Integrity Research Institute, a non-profit organization dedicated to energy research and public education.





Secrets of Nicola Tesla's Experiments

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At the close of the 19th century the great scientist Nicola Tesla gave to the whole world a demonstration of electrical power transmission using a single incomplete and unearthed wire. Nevertheless, the nature of this phenomenon

remains obscure till nowadays. It is also known that Engineer Stanislav Avramenko succeeded to repeat the famous experiment. But as far as we know, mention of the physical nature of this phenomenon has been never made. The present article makes an attempt to explain how “it” works.

At the dawn of electrical science there occurred the idea of existence of a certain electrical liquid, which is able under certain conditions to flow from one body to another and to be both in abundance and in shortage. B. Franklin introduced at his time the notion of positive and negative electricity. J.C. Maxwell used in his theoretical investigations the direct analogy between water movement and electricity movement.

It is common knowledge that the electrical current is any movement of electrons (in this particular case – their movement in a metal), which happens when there is the potential difference. Is there an explanation of the movement of electrons in a single conductor?

As an example let us take an ordinary garden hose with some water inside it. Both ends of the hose are closed with plugs. How to make the liquid move? The only way is to start rotation of the liquid from one end so that to transmit its rotation to the other end of the hose. To make the liquid “move” it is necessary to establish an alternating current in the hose by moving the liquid

alternatively from one end to another. But even in this case the water in the hose will not be moving the way we need. So, having pulled the plugs out, let us try to attach a vessel to each end of the hose (let the vessels have the form of a cylinder).

If we mount a plunger in one of the vessels and keep pushing it down we will make water in the hose move from the first vessel to the farther one (communicating vessels). Then if we pull the plunger up, as a consequence of wetting the plunger (water adheres to the plunger), water from the second vessel will flow through the hose back to the pump vessel.

If we continue the described manipulation then a water current of alternating direction will arise in the hose. In case if we manage to mount a screw with vanes (a propeller) inside the hose (in any of its point), it will rotate alternatively to both directions thereby proving the fact that moving liquid carries energy in it. A similar process takes place when we use a single wire.

As you know, the electroscope is an elementary device for detecting the presence of an electric charge. The simplest model of it consists of a glass jar with a plastic cap (an insulator). A metal stick is inserted into the jar through a hole in the middle of the cap. Over the cap there is a ball made of the same material as the stick itself.

There are lightweight strips of foil attached onto the lower end of the stick inside the jar and hanging one opposite to another. The strips can easily move attracting or repulsing each other. Let us remember that if we rub an ebony stick with a patch of wool, the stick will get charged. Then if we approach it to the upper part of the electroscope (to the ball) the foil strips in the jar will immediately move apart forming a certain angle and thereby proving that the electroscope has got charged.

When this procedure is finished, let us place another uncharged (with drooping strips) electroscope within the distance of three meters from the first one. Let us connect both electroscopes with a bared wire, holding its insulated middle part with our fingers. At the moment when the wire touches the upper balls of the both electroscopes we will see that the second (initially uncharged) electroscope will immediately “come alive” – its strips will set apart (forming a smaller angle than it was initially in the first electroscope) and the strips of the first electroscope will go a little down. Now the both electroscopes show the presence of charges that have been transmitted from the ball-capacity of the first electroscope to the ball-capacity of the second one.

The charges of the both electroscopes have equalized. It is clear that in this case electrons have flown from one electroscope to another and instantaneous current has arisen in the wire. If we obtain first charging and then discharging of the first electroscope from one end in constant duty, it will be evident that an electrical current of alternating direction will run in the wire between the electroscopes. Let us mention that the first electroscope is to be charged with one sign and discharged with the opposite one.

You can find a description of this process in any of full-length physics textbooks. But they never mention that it is possible to make the process constant or how it can be applied either. It seems to be strange enough that such a problem can bring many investigators to a nonplus.

To develop the subject, let us note that we can obtain the same continuous process (i.e. single wire excitation of alternating current) using the well-known method of electrostatic induction (field effect). For that it is enough to excite a near

ball or sphere with a charged body, e.g. with a rubbed ebony stick. This effect must be alternating: sometimes by approaching, sometimes by taking the stick aside without touching the ball.

Theoretically nothing will change if we rotate (e.g. using a small motor) two electret balls of unlike charges located diametrically at the near sphere or ball. A current will run in the conductor from the near ball-capacity to the farther one and back. It is viable to use an electrostatic machine (that will divide and accumulate charges of the opposite sign) or an electrostatic generator with line supply that will play the same part. In case if we supply alternately plus and minus charges from the electrostatic generator to the near sphere (we may obtain such alternation by using two relays or semiconductor switches), then electrons in the wire will run from the distant ball-capacity at supplying a plus charge and they will run back at a minus charge. We should mention that when potential difference arises in the wire then the strength of electrical field in our process becomes constant. Now when electrons have a place where they can flow to (to the balls-capacities) we may excite alternating current by applying the method of electromagnetic induction. In other words if a part of the conductor is coiled then forcing it dynamically with a magnet will entail the same result. Thus for this purpose one might use a transformer as well. The current can also result from exciting opposite ball-capacities in turns, i.e. from the both sides. In order to obtain a big potential of the ball-capacities (by their direct charging or using the method of electrostatic induction) we may apply a well-known principle of Van der Waals generator. By means of such a generator it is possible to obtain a potential of millions of volts and, consequently, rather high voltage.

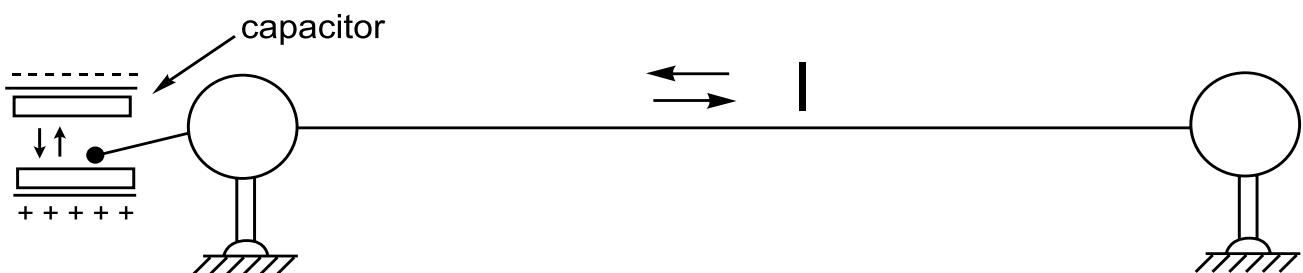


Fig. 1

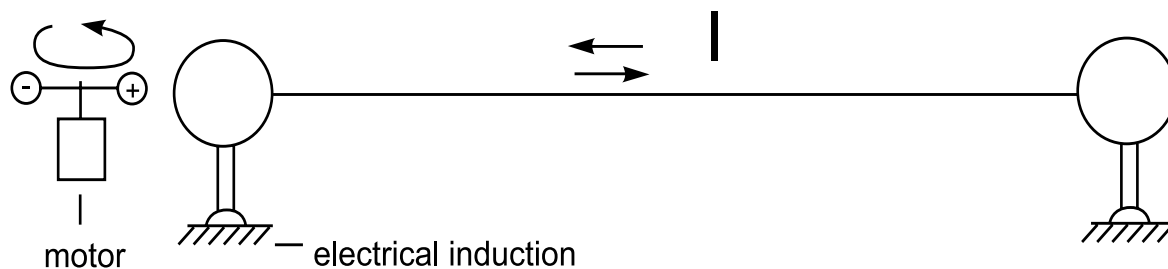


Fig. 2

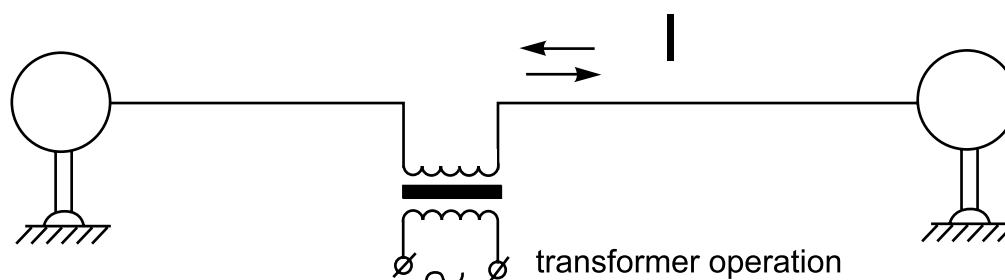


Fig. 3

In addition let us remind that lightning strikes sometimes from clouds (from above), sometimes upward from the earth and sometimes between thunderstorm clouds. This is one more indirect proof of the fact that it viable to transfer AC using a conductor. Besides it is also worth mentioning that alternating current can be transformed into direct current.

So if we install appropriate (new) generators to electrical power plants then old electrical power transmission lines will be able to transmit higher wattages than they transmit today because transmission of the same amount of power will require less quantity of wires (the rest of the wires will be released).

Using the abovementioned method it is possible to transmit electrical energy in the form of excitation of an electrical field from “this” side to any distant point of the planet. It is related with that the Earth is a large conductive globe (and a charged one), besides charges can be divided – polarized to opposite ones.

On the whole by receiving an initial signal by means of an appropriate receiver of the antipodal point we have obtained not only a way of power transfer, but also a way of information transmission. As we modulate a signal in one point, in the other point it is demodulated.

Besides, the principle of modulation-demodulation can be applied to single-wire transmission as well. It should be noted that energy and information transmission is viable in case if we induce the magnetic field of the planet from “this” point”.

We will not dwell on the “torsion” principle of single wire transmission of electrical energy (in this case both the electrical field and, correspondingly, electrons rotate from one end to another transmitting this rotation in a wire to its opposite end). Maximal length of the wire depends on the amount of a potential on the ball-capacity. The capacity itself depends on its own radius.

Let us consider some questions that N. Tesla probably never investigated. A hypothesis that may be working – that is to say a viable one – is presented below. The author carried out the following experiment. A cylindrical permanent magnet was suspended on a thread. When the magnet set still, another analogous magnet was carried to it at a certain distance. The second magnet was oriented towards the first one with its opposite pole.

At that there was observed a certain deviation of the first magnet. The first suspended magnet was fixed from both sides. So the magnet could move

only in one plane, strictly within the arch that depended on the radius of the suspension. Then the experimenter struck the field of the second intermediate stationary magnet (all the magnets were oriented one to another with opposite poles) abruptly with the field of a third magnet. As a result of it the first magnet, which was located on the other side of the intermediate stationary magnet, also bounced abruptly aside.

Most probably, there was a pulse transmitted in the magnetic fields of these interacting magnets. It reminds another well-known effect: if we place ten contacting balls in one line on a horizontal flat surface and hit the outermost ball, then nine balls stand still and the last ball on the opposite end rebounds.

If it is possible in case with the balls, then why should it be impossible with a row of magnets that are oriented with opposite poles one to another and are rigidly fixed to a flexible tube? If we run energy through this new "wire" by preliminarily exciting one of its ends with an

abrupt pulse of the magnetic field, this energy may be received on the other end of the conductor with a magnetic field receiver. If we take a solid metal wire and magnetize it so that the field lines are oriented parallel to the field axe, in this case we will also obtain a new conductor able to play the mentioned role, that is to transmit a pulse through the magnetic field of the "wire" from its one end to another.

We can apply the same principle to like charged balls or more precisely to electret (like) balls or to an electret (solid) wire. But in this case we will have to "strike" with an electrical field from one side so that to transmit a pulse to the other end.

The author believes that realization of the idea described above will entail creation of a new generation of technical devices and that single wire transmission of non-mechanical energy is viable.

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Longitudinal Waves

Cold Electricity

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Dr. Peter Lindemann in his book "The Free Energy Secrets of Cold Electricity" presents the story of this type of electricity which really is not electricity as electricity is normally meant to be understood. Cold electricity behaves differently than normal electricity. Regular electricity may be used in the process to generate this cold form but this transformation is not the type of change in voltage and current that occurs in a transformer. Rather, it is the extraction of a form of current from "normal" electricity by a process that uses high voltage.

In discussing the Edwin Gray motor, Dr. Lindemann relates a demonstration by Gray. Gray used a 6 volt car battery with lead wires running to a system he had devised using a voltage booster and a series of capacitors which raised the voltage to 3,000 volts. He then closed a switch that ran the high voltage into two electromagnets which caused a loud popping sound and cause the top electromagnet weighing a pound and a quarter to be propelled over two feet into the air. Gray claimed that only 1% of the system's energy was used with 99% going back to the battery. He said that he had 'split the positive' with his system.

Other evidence of using this fractionated electricity was Gray's use of a small motorcycle battery (15 amps which would normally produce $W = V \times A = 6 \text{ V} \times 15 \text{ A} = 90 \text{ watts}$) to run concurrently six 15-watt electrical light bulbs, a portable 110 volt T.V. set and two radios. A glowing 40 watt light bulb running off the system was dropped into water without the glass breaking; the bulb gave off light but not the heat that accompanied a bulb run by regular electricity.

This would mean that tungsten in the filament of the light bulb was not resisting the flow of this fractionated electricity and that this "cold electricity" was causing the luminous glow by some other means.

According to Gray's Patent #4,595,975, low voltage was chopped into pulsating DC by passing it through a multi-vibrator (buzzer like a door bell). Then the pulsed DC was sent through the low voltage winding (primary) of a transformer which changed it to pulsating high voltage DC current at the secondary side. The pulsating high voltage DC was rectified by a full wave bridge and changed into high voltage DC.

The high voltage DC was used to repeatedly charge a capacitor as the current was sent to briefly discharge across a spark gap (rated at 3000 volts). The discharge of the spark across the gap had to be in one direction only and its duration was controlled by the size of the capacitor and by the strength of a magnetic field encompassing the gap. This magnetic field had a quenching effect because it caused a back EMF each time the discharge occurred. The current from the discharge across the gap was then sent through a resistor and then to a vacuum tube (the conversion switching element tube).

Another of Edward Gray's patents "Efficient Electrical Conversion Switching Tube Suitable for Inductive Loads" (April 1987, patent 4,661,747) described the switching tube. It utilizes a low voltage anode (positive plate), a low voltage anode (positive plate) and one or more electro-static or charge receiving grids (located between the positive

plates and the cathode or negative plate supplying the electrons). This tube's function was to 'split the positive'. A normal vacuum tube usually uses only a single anode at a particular voltage to control the voltage flow within the tube. The function of the charge-receiving grids was to collect the "cold electricity".

The most unusual part of this Power Supply circuit consisted of devices specifically designed to remove excess energy when the Switching Tube was functioning. A spark-gap protection device, served to "protect the inductive load and the rectifier elements from unduly large discharge currents". As stated, the rectifier elements are the two anodes and the grids in the Switching Tube. In a normal vacuum tube, no protection would be necessary because the maximum power of the tube would be based upon the electricity provided to heat the tube filament and to charge the anode (voltage x amps). Therefore, the excess energy comes from the process occurring as the spark gap affected electrical pulses pass through the Switching Tube. The process of extracting the "cold electricity" places danger of overload upon the system. (Please take note that the "electricity" from the Switching Tube is no longer the normal type of electricity we have from our batteries or from an electrical outlet but rather a completely new form with entirely different characteristics.)

The description continues, "Should the potentials (voltages) within the circuit exceed the predetermined values fixed by the mechanical size and spacing of the elements within the switching tube, the energy is dissipated (bypassed) by the protective device to the circuit common (electrical ground)" by two strategically placed diodes (devices that permit the passage of electricity in one direction only). The question now arises "What is the cause of this large amount of excess energy which must be dumped to ground through device acting like a lightning rod in order to prevent the circuit from burning out?"

To examine the cause of the excess energy source, we must go back over a hundred years to 1889. Heinrich Hertz had just announced

in 1887 that he had discovered electromagnetic waves and Nicola Tesla was attempting to duplicate Hertz's experiments. Tesla used abrupt and powerful electric discharges produced with banks of capacitors charged to very high potentials and was able to explode thin wire (copper bus bars). He came to the conclusion that Hertz had mistakenly associated electrostatic inductions (electrified shockwaves in air) for true electromagnetic waves.

The explosion of the copper bars by means of the "disruptive discharges" from the capacitor bank produced sharp shockwaves which struck Tesla with great force across the entire front of his body. Tesla said they felt more like powerful gunshots rather than electrical sparks. They produced effects similar to lightning or to those produced by high voltage DC generators previously discussed; the simple closing of a high voltage DC generator caused a stinging shock. AC generators were not in use in this pre-AC era and it was shown, later on, that AC generators did not produce these effects.

This shock was first assumed to be the result of residual static charging. It stood straight out of highly electrified conductors, seeking ground paths which included workmen and switchboard operators. In long cables, Tesla estimated that this electrostatic concentration was several orders in magnitude greater than any voltage the DC generator could actually produce. It caused a hedge or crown of bluish needles or spicules to point at right angles to the cable or straight from the electrical cable line into the surrounding space. The bluish needles appeared the very instant the switch was closed and disappeared a few milliseconds later, after which the system functioned normally. However, anyone through whom the blue needles passed, especially in large regional power systems which used high exceptionally high voltages, usually did not survive. Generators rated at a few thousand volts produced hundreds of thousands or even millions of electrostatic volts during the start-up pulse. Highly insulated, heavily grounded relay switches had to be installed to protect workers from certain death.

Engineers at this time assumed **the effect to be caused by a "bunching" action which occurred when a powerful force was not able to move charge sufficiently quickly through a system.** (A similar "choking effect" also happened with large steam engines; if steam was introduced too rapidly, the steam engine could explode.) The metal of which the wire was composed somehow offered resistance to the charge carriers before they were able to move away from the generator terminals. It seemed that the wire acted as a blocking force to the electrons or to some unknown part of the electrons for the first few milliseconds, rather than as a conductor. The powerful, deadly bluish spikes sprang from the line until the current charges actually "caught up" with the applied electrical field. It was as if the spicules were a form of relieving pressure on the system by converting the voltage impulse into some other form.

Tesla began wondering why the electrostatic field moved faster than the actual charges. He believed that this effect could help him find electric waves better than his capacitors because the line resistance caused the electrostatic charges to bunch together into a density much greater than obtainable with his capacitors.

Tesla understood that ordinary capacitor discharges were oscillating currents or spark currents which "bounced" between each capacitor plate until their stored energy was dissipated. The high surge voltage of the DC generator exerted such a high one way pressure on the dense line charges, due to the millisecond resistance of the metal, that possible voltage "bounces" were also oscillations of current as was true with the capacitors. Tesla used every means possible to block the "back-rush" (bouncing) of the oscillating current in order to prevent the supercharge to prematurely decline; he wanted to maintain this powerful "bunching" effect as long as possible in order to study it and to make use of it.

His face and hands experienced a penetrating shockwave; a sharp pressure and an electrical irritation were brought on at the sudden closing of the switch. His face and hands were

especially sensitive to these shockwaves which caused a "stinging" effect at close range. Tesla believed that he was being struck by material particles approaching the vapor state as they were thrust from bluish crowned wire.

Later on, he was to discover they were not gas particles. **He placed himself behind a glass shield but to his amazement, he still felt the shockwaves and stinging effects; the glass shield did not protect him.** These stinging rays could be felt at great distances from their sources and, to his amazement, would penetrate shields of both glass and copper. Tesla knew that normal electrostatic charges spread over the surface of a metal (copper, for example) shield so that the stinging effect could not have an electrostatic origin and therefore was not electrostatic in nature but was from some other yet unknown source.

Joseph Henry in 1842 had noticed the magnetization of steel needles by the discharge of a Leyden Jar (type of primitive condenser made with a glass bottle). The Leyden Jar was at the top floor of a building while the needles were in the basement. Dr. Henry, noting the passage of these magnetizing rays through brick walls, oak doors, stone and iron flooring and tin ceilings believed that the spark given off by the discharge released "light-like rays" that passed through the material obstacles with ease before magnetizing the needles.

Elihu Thomson, a physics instructor in 1872 had been attempting to make sparks from a Ruhmkorff Spark Coil more clearly visible for his physics students. He attached one pole of the coil to a cold water pipe and noted that the previously blue spark changed to white. He then attached the other pole to a large metal table top and produced a silver-white spark that would be clearly visible to all attending his lecture. He went to the door of the room to notify his colleague but received a strong shock from the door knob. Turning off the Ruhmkorff coil prevented the brass knob on the oak door to stop shocking. He returned with his friend, again turned on the coil and discovered that all metal objects in the entire building, no matter how distant

from the coil or insulated from the floor, produced long and continuous white sparks upon the touch of a penknife or screwdriver.

The device Tesla perfected was far superior to the Ruhmkorff Spark Coil. The effect of his disrupter device was so strong that a single wire placed in an oil bath produced what he had originally but mistakenly thought to be gaseous streams so powerful that they visibly depressed the oil into a depth of about two inches. Tesla concluded that "besides the air, another medium is present".

...he placed himself behind a glass shield but to his amazement, he still felt the shockwaves and stinging effects; the glass shield did not protect him. These stinging rays could be felt at great distances from their sources and, to his amazement, would penetrate shields of both glass and copper...

Tesla had been able to perfect his device by placing a capacitor between the switch and the D.C. generator thus increasing the power and protecting the generator windings in much the same manner as the diodes Edwin Gray's Switching Tube protected Gray's circuit. Tesla also raised the voltage and quickened the "make-break" rate of the switch to increase the power level by placing a powerful permanent magnet crosswise to the discharge path of the high voltage output of the DC generator wires. The magnetic field caused the discharge arc to automatically "blow out"; the charge passing through the wires in the presence of the magnetic field built up reverse emf (electromotive force or voltage).

Tesla believed he had discovered a new type of electricity with special characteristics. This form of electricity did not consist of alternating waves. **They were longitudinal waves** composed of successive shocking

waves with effects that could be seen and felt at a distance. **Vector components of these shock waves were unidirectional. They were able to force charges in the direction of their propagation.** In his patent #787,412, "Art Of Transmitting Electrical Energy Through The Natural Mediums" (April 18, 1905), Tesla calculated the mean velocity of the waves propagated by his device to be 471,240 Km/sec. Knowing that the velocity of light is 300,000 Km/sec would indicate that the type of transmission to which Tesla is referring is different than standard electromagnetic radiation; these special longitudinal "Radiant Energy" waves moved faster than the speed of light.

After conducting hundreds of experiments, Tesla found that "Radiant Energy" longitudinal waves **could penetrate all materials and cause "electronic responses" in metals such as copper and silver.** Impulses exceeding 0.1 millisecond duration produced effects such as pain, mechanical pressures, explosion of thin wires and vibration of objects. Impulses of 1.0 microseconds caused the sensation of physiological heat; at still shorter lengths white light would fill the room. Impulses less than 100 microseconds were safe to work with and Tesla planned to use them for his power broadcast system since they could pass through all matter.

In 1890, Tesla discovered that placing a long single turn copper helix near his magnetic disrupter became covered with an envelope of white sparks. Effects were strongest when the helical coil was placed within the disrupter wire circle. In this "shockzone", the coil was surrounded by a blast of long, fluid like silvery streamers which clung to the surface of the coil, flowing over the coil at right angles to the windings. Tesla hypothesized the electrostatic-like effect was due to radiant transformation rules requiring measurements of discharge lengths and attributes of the helix used (number of turns, diameter, etc.). (Editor: resonance)

The new induction law he discovered showed that radiant shockwaves became much stronger when encountering segmented objects. The radiant shockwaves

"flashed over" the outer skin of the helix from end to end without passing through the windings of the coil; shockwaves of 10,000 volts introduced to a 24 inch coil rise to 240,000 volts. The greater the resistance in the helix turns, the higher the maximum voltage. This was completely different from magneto-induction. Remember that he was not working with electricity but with radiant shockwaves.

Aether particles were incompressible and could easily pass through space and matter with a speed much greater than the speed of light. This was pure radiant matter but, at the same time, pure Radiant Energy. Cold electricity is one of the forms of this Radiant Energy

This transformer he invented used radiant shockwaves to produce pure voltage without current. Each transformer had to be "tuned" by adjusting the disrupter to a specific impulse duration. At that point, voltage impulses could flow smoothly, flowing over the copper surface much like a stream of water in a pipe. No amperage could be detected but if the stream was aimed at a distant metal plates, "current" was produced which reached several hundred or even thousands of amperes. Tesla then began to wonder what made up this white, currentless stream.

Tesla determined that normal charge carriers (electrons) could not travel as quickly as the radiant pulse; no current moved the coil because the electrons were choked in the metal lattice of the coil. **The radiant pulse moving over the surface of the coil was not electronic in nature.** He placed the legs of a heavy U-shaped copper bus bar directly to the disrupter primary and then connected the short-circuited system to several incandescent lamps. These lamps glowed with a brilliant cold white light as

with Gray's light bulb, proving that the power for the light was not electrical in nature. Tesla believed that the electrons were blocked from flowing through the wire while the radiant pulse was released over the coil surface as a "gaseous" pulse (splitting the positive, according to Gray).

Tesla came to believe that voltage could be viewed as streams of aether under various states of pressure and that his transformers affected the aether as to produce the luminous effects he observed. In his patents, he describes his "light-like rays" as tightly constricted aether streams propelled from his transformers along infinitesimal ray lines along which an incompressible movement occurs instantly through space on all points along its path. Aether streams were being drawn in through his transformer at higher natural pressure and then accelerated in the electrical discharge. The voltage in his transformer could control the brilliance of light in a room but this type of light was almost impossible to register on film. He could heat up a room or cause cool breezes by controlling the voltage in impulse duration in his transformer.

Tesla saw electrical current was really a complex combination of aether and electrons. Through the application of a disrupter, the electrons were removed from the gap by the magnetic field while the aether streams continued to flow through the circuit. He considered aether particles highly mobile with infinitesimally small mass and cross-section as compared to electrons. They were incompressible and could easily pass through space and matter with a speed much greater than the speed of light. This was pure radiant matter but, at the same time, pure Radiant Energy. Cold electricity is one of the forms of this Radiant Energy.

Edwin Gray, Dr. Nicola Tesla and Dr. Thomas Henry Moray all used radiant energy technologies. It is important to understand that the Laws of Thermodynamics and Maxwell's equations do not pertain to "Radiant Energy" technologies.